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Machine learning advancements in organic synthesis: A focused exploration of artificial intelligence applications in chemistry

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ABSTRACT

Artificial intelligence (AI) is driving a revolution in chemistry, reshaping the landscape of molecular design. This review explores AI's pivotal roles in the field of organic synthesis applications. AI accurately predicts reaction outcomes, controls chemical selectivity, simplifies synthesis planning, accelerates catalyst discovery, and fuels material innovation and so on. It seamlessly integrates data-driven algorithms with chemical intuition to redefine molecular design. As AI chemistry advances, it promises accelerated research, sustainability, and innovative solutions to chemistry's pressing challenges. The fusion of AI and chemistry is poised to shape the field's future profoundly, offering new horizons in precision and efficiency. This review encapsulates the transformation of AI in chemistry, marking a pivotal moment where algorithms and data converge to revolutionize the world of molecules.

1. Introduction

In the era of digital chemistry, automation, robotics, and artificial intelligence (AI), a profound revolution in the field of chemistry is being witnessed [1–3]. The research domain is currently referred to as digital chemistry, encompassing the integration of advanced automation tools, machine learning algorithms, and the application of big data [4,5]. Its objective is to combine digital technology with traditional experimental chemistry to expedite the discovery and synthesis of novel compounds [6–8]. By leveraging digital tools to create innovative workflows, chemistry is advancing at a faster, more cost-effective, and sustainable pace [9]. This transformation not only significantly enhances productivity but also fundamentally alters how chemists approach their work and innovation [10]. It alleviates the burden of labor-intensive experimental work and aids chemists in gaining deeper insights into reaction mechanisms. Within this transformative landscape, one of the most noteworthy applications of AI is observed in the realm of organic synthesis [11].

Given the complexity and non-linear nature of organic chemistry, it is understood that efficiently performing various tasks, such as retrosynthetic analysis, designing new chemical reactions, and optimizing reaction conditions, requires years of expertise and specialized chemical knowledge [12]. However, there is a growing expectation that these processes can be streamlined and automated, at least partially or in highly controlled environments, to enhance productivity and ensure the reproducibility of results, especially when coupled with robotic systems [13]. Indeed, in the field of organic synthesis, the impact of AI has been particularly profound. AI plays a pivotal role in retrosynthetic analysis, involving the reverse engineering of a series of reaction steps to synthesize a target molecule starting from existing compounds [14]. Traditionally, this has been a labor-intensive and knowledge-driven task. However, machine learning algorithms are now capable of generating synthetic pathways based on known reaction data and rules [15]. This enables chemists to rapidly design new synthetic routes, expediting the discovery of novel drugs and materials. Another significant

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application lies in reaction prediction, where AI algorithms forecast the products of chemical reactions by analyzing reaction conditions and reactants [16]. Organic synthesis encompasses a vast number of reactions, traditionally necessitating extensive experimentation to determine optimal reaction conditions. However, machine learning algorithms can identify patterns from extensive reaction datasets, enabling the prediction of novel reaction outcomes [17]. This not only saves time and resources but also facilitates the discovery of new reaction conditions [18]. Furthermore, AI can be employed for the optimization of reaction conditions, maximizing reaction yields and selectivity [19]. By analyzing a wealth of experimental data, machine learning algorithms can identify optimal reaction conditions, assisting chemists in designing experiments more effectively [20].

Despite the remarkable progress achieved by digital chemistry in the field of organic synthesis, several challenges remain: The quality and availability of data are of paramount importance. Machine learning algorithms rely heavily on high-quality data for accurate predictions and optimizations. For this, high-throughput synthesis (HTS) either manual or automated, can enhance the efficiency of ML by providing large, diverse datasets of reaction outcomes and compound properties. The abundance of data from HTS enables more accurate predictions, better model generalization, and improved performance in guiding future experiments and optimizing reaction conditions. The integration of AI and robotics can further increase the efficiency of HTS to produce more accurate and robust data for ML. Hence a continuous improvement cycle can be triggered which can lead to new discoveries. Therefore, the collection, curation, and sharing of data become essential [21].

Many AI algorithms, particularly deep learning models, are often considered black-box models, making it challenging to interpret their internal decision-making processes. This may raise issues of trust and safety in chemical research, as chemists need to understand why specific predictions or recommendations are made [22]. AI algorithms are data-driven and may lack profound domain expertise. In chemistry research, a deep understanding of chemical reaction mechanisms and complexity is crucial, and AI algorithms may fall short in this regard [23]. Handling uncertainty is a persistent challenge, especially in complex chemical systems where reactions and molecular interactions are subject to various factors, including temperature, pressure, and solvents. Some advanced AI models require substantial computational resources and specialized hardware for operation.

Digital chemistry and artificial intelligence are ushering in unprecedented opportunities and challenges in the domain of organic synthesis [24]. With ongoing technological advancements and the dedication of chemists, further breakthroughs can be anticipated, fostering innovation in both scientific research and industrial applications [25]. In this review, we will primarily focus on the extensive applications of AI in organic synthesis, illustrating the promising future prospects of AI in chemistry through specific examples.

2. Machine learning approaches

Machine learning, a dynamic field at the intersection of computer science and statistics, encompasses a range of learning approaches that empower systems to autonomously improve their performance based on data. These approaches can be broadly categorized into three main paradigms (Fig. 1).

2.1. *Supervised learning*

Supervised learning is a type of machine learning where the algorithm learns from labeled data. Labeled data means that the input data is accompanied by the correct output or target value. The algorithm is trained to learn the relationship between the input and the desired output and make predictions on new, unseen data. It is commonly used for tasks such as classification, where the algorithm learns to classify data into different categories, and regression, where the algorithm learns to predict continuous values [26].

2.2. *Unsupervised learning*

Unsupervised learning is a type of machine learning where the algorithm learns from unlabeled data. Unlabeled data means that the input data is not accompanied by any target value or output. The algorithm explores the structure or patterns in the data and learns to make sense of it without any specific guidance. It can be used for tasks such as clustering, where the algorithm groups similar data points together, and dimensionality reduction, where the algorithm reduces the number of features or variables in the data [26].

2.3. *Reinforcement learning*

Reinforcement learning involves an agent interacting with an environment and learning to take actions to maximize cumulative rewards over time. Through a trial-and-error process, the agent learns optimal strategies, making it suitable for long term applications. An essential hurdle in reinforcement learning revolves around the exploit-explore dilemma, which entails striking the right equilibrium between selecting actions with established high expected rewards and delving into actions that remain unexplored [27].

Additionally, these primary paradigms often intersect, giving rise to hybrid approaches like semi-supervised learning, which combines labeled and unlabeled data, and transfer learning, which leverages knowledge from one task to improve performance on another. Machine learning encompasses a variety of algorithms tailored to each learning method, allowing systems to acquire knowledge and improve performance through data analysis. Here's a breakdown of algorithms commonly used in each learning method.

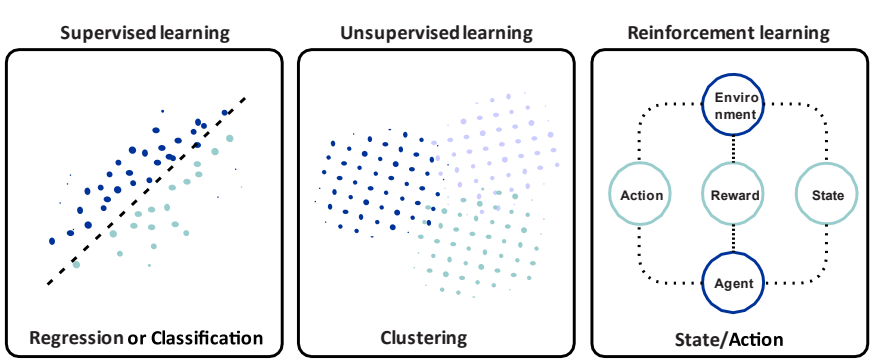


Fig. 1. Machine learning approaches.

3. Machine learning pathway

The process of applying machine learning in chemistry is a systematic journey that merges data analysis, computational algorithms, and domain expertise to accelerate the discovery and synthesis of new molecules. This intricate process unfolds through different essential stages, each contributing to the goal of identifying desired outcomes (Fig. 2).

3.1. Data Collection and Preprocessing

Machine learning in chemistry begins by gathering a wide array of chemical and biological data. This raw data is then refined and transformed into meaningful features that can be effectively used for analysis.

3.2. Model Selection, Training, and Tuning

The process involves selecting the right machine learning algorithms that suit the task at hand. These algorithms are trained using the available data, and their performance is enhanced through careful tuning of parameters for optimal results.

3.3. Validation, Testing, and Interpretation

To ensure the reliability of the models, validation, and testing are conducted using new and unseen data. The insights gained from these models are used to interpret the relationships between the features and the properties being predicted.

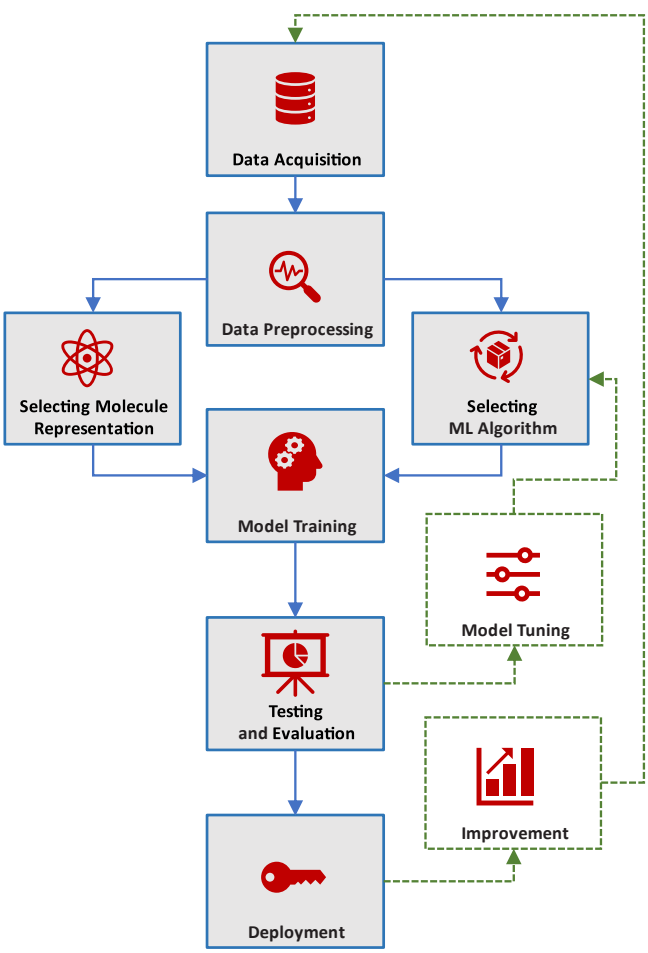


Fig. 2. Simplified machine learning process.

3.4. Virtual Screening and Deployment

The validated models are deployed for virtual screening, allowing efficient screening of a vast number of compounds. This accelerates compound selection and optimization for further experimentation and testing.

3.5. Continuous Monitoring and Improvement

To maintain the accuracy and relevance of the models, continuous monitoring is essential. The models are updated and improved as new data becomes available, ensuring their reliability over time. (Table 1).

4. Representation of molecules for machine learning

The representation of organic molecules in machine learning is a crucial step with far-reaching implications. It directly affects prediction accuracy, computational efficiency, and model interpretability. A well-chosen representation enhances generalization, facilitates data utilization, and offers valuable chemical insights. Overall, this step significantly influences the quality and applicability of machine learning models in chemistry, materials science, and drug discovery. There are various ways to represent molecules in the context of machine learning and computational chemistry. Some common methods are listed in Table 2.

Beside the descriptors introduced in Table 2, there are some particular descriptors used for reaction prediction and optimization studies including dragon descriptors (geometrical descriptors, 3D matrix-based descriptors, 3D autocorrelations, RDF descriptors, 3D MoRSE descriptors, WHIM descriptors, GETAWAY descriptors, Randic molecular profiles, 3D atom pairs, CATS 3D descriptors), molecular operating environment (MOE) descriptors, average steric occupancy (ASO) descriptors, extended connectivity fingerprints (ECFP6) descriptors, and Mol2vec descriptors [47].

The selection of descriptors in the context of conveying physicochemical information in input data is acknowledged as a crucial yet intricate task. However, the process of descriptor selection demands a profound understanding of the specific problem settings, often held by specialists well-versed in the domain. Yet, it is imperative to note that the computational cost associated with computing numerous descriptors can be substantial, potentially impacting the time required for inference. Moreover, the validity of descriptors may be confined to specific regions within the vast chemical space, posing a challenge in achieving a universally applicable and accurate representation [48]. For example, there have been proposed many ML models for prediction of aqueous solubility of organic molecules trained on experimental data. But due to the variance in the chemical space of covered compounds, measurement methods, experimental conditions, and non-standard representations of molecules in existing solubility datasets, there is vast difference in prediction of different models. Models based on string representations, 3D descriptors or 2D descriptors exhibited different degrees of accuracy [49–51].

Various computational tools and libraries are used to extract features or different descriptors from molecules. Some commonly used tools for extracting descriptors are listed in Table 3. In following section, we will discuss ML approaches used for optimization or prediction of various tasks in organic chemistry.

5. Yield prediction

The utilization of machine learning models in predicting reaction yields through machine learning architectures has gained significant attention. This approach involves the integration of various analytical techniques to optimize the accuracy of yield forecasts.

In one of the early attempts the Doyle group compared several ML models for yield prediction in palladium-catalyzed Buchwald-Hartwig amination reaction (Fig. 3) [71]. They evaluated linear regression

Table 1
Molecular representation in machine learning.

Supervised Learning Models	
Naïve Bayes	Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem. It assumes feature independence given the class label, making it efficient for high-dimensional data[28].
Support Vector Machines (SVM)	SVM is a discriminative classifier that finds an optimal hyperplane to categorize new examples. The hyperplane is defined by support vectors, which are the data points closest to the decision boundary[29].
Random Forest (RF)	RF is an ensemble method that combines multiple decision trees to make predictions. It uses a voting system for classification problems and averaging for regression problems. RF helps prevent overfitting by combining multiple decision trees[30].
K-Nearest Neighbors (KNN)	KNN is a nonparametric algorithm that determines the class of a data point based on the similarity of its neighbors. It assumes that similar data points exist in close proximity and assigns the majority class of its k nearest neighbors[31].
Artificial Neural Networks (ANN)	ANN are composed of interconnected layers of neurons. They learn from input data and perform classification or regression tasks. ANNs can be trained through supervised learning by adjusting the connections between neurons to minimize the error[32].
Unsupervised Learning Models	
K-means clustering	K-means clustering is a popular unsupervised machine learning algorithm used for partitioning a dataset into a predetermined number of clusters. The goal of K-means clustering is to group similar data points together while keeping dissimilar points in separate clusters. It is an iterative algorithm that assigns data points to the nearest cluster center and then updates the cluster centers based on the mean of the data points in each cluster. This process continues until the algorithm converges, and the cluster assignments stabilize[33].
Fuzzy clustering	Fuzzy clustering is a variation of traditional clustering methods like K-means that allows data points to belong to multiple clusters with varying degrees of membership rather than being exclusively assigned to a single cluster. It assigns each data point membership values across all clusters, reflecting the likelihood of belonging to each cluster. This soft assignment enables a more nuanced representation of complex data relationships, making fuzzy clustering valuable in scenarios where data points have ambiguous cluster affiliations or when there is inherent uncertainty in the data[34].
Hierarchical clustering	Hierarchical clustering arranges data points into a tree-like structure, merging or splitting them based on similarities to form nested clusters. It offers a flexible way to visualize data relationships across different levels of granularity, making it useful for exploring data structure without specifying the number of clusters beforehand[35].
Principal Component Analysis (PCA)	PCA is a method to reduce the complexity of high-dimensional data by identifying key patterns and directions of variation. It transforms the data into a lower-dimensional space while retaining important information, making it beneficial for visualization and analysis of large datasets[36].
Independent Component Analysis (ICA)	ICA is a computational technique that separates a multivariate signal into additive, independent components, assuming that the observed data is a linear combination of these sources. Unlike methods like PCA, which aim to capture orthogonal directions of variation, ICA aims to extract statistically independent components, useful for tasks like blind source separation and signal denoising. It has applications in fields like image processing, neuroscience, and telecommunications, where the goal is to uncover hidden sources from mixed observations[37].
Autoencoders	Autoencoders are neural network architectures, that consist of an encoder that maps input data into a lower-dimensional latent space representation and a decoder that reconstructs the original data from this representation. By forcing the network to learn a compact representation of the input data, autoencoders can capture meaningful features and reduce noise. They find application in tasks such as dimensionality reduction, data denoising, and anomaly detection, as well as generating new data samples similar to the input data distribution [38].
Deep Belief Nets (DBNs)	DBNs are complex generative models composed of multiple layers of interconnected nodes, where each layer is trained to model higher-level abstractions of the data. DBNs consist of a stack of restricted Boltzmann machines (RBMs) or similar components, allowing them to learn hierarchical representations of input data. DBNs are often used for feature learning and unsupervised pretraining in deep learning tasks, and they excel in tasks involving complex data structures such as images or sequences. They have been used in various domains including computer vision, natural language processing, and recommendation systems[39].
Generative Adversarial Networks (GANs)	GANs are deep learning models consisting of two neural networks, a generator, and a discriminator, engaged in a competitive process. The generator creates fake data samples, while the discriminator evaluates whether samples are real or fake. Through iterative training, the generator improves its ability to create increasingly realistic data, while the discriminator refines its ability to distinguish real from fake. GANs have revolutionized generative modeling, enabling tasks such as image synthesis, style transfer, and data augmentation. They have also spurred advancements in unsupervised learning and adversarial training paradigms[40].
Self-Organizing Map (SOM)	SOM, also known as Kohonen map, is an unsupervised neural network technique used for dimensionality reduction and visualization of high-dimensional data. It involves a grid of nodes or neurons that represent a lower-dimensional space. During training, data samples are presented to the SOM, and each node competes to be activated by finding the most similar data samples. Over time, neighboring nodes become more alike, causing similar data to cluster together on the map. SOMs are useful for tasks such as clustering, data visualization, and understanding the underlying structure of complex datasets[41].
Reinforcement Learning (RL) Models	
Q-Learning	Q-Learning is a fundamental reinforcement learning algorithm that learns optimal action-selection strategies by estimating Q-values, representing expected cumulative rewards. It iteratively updates Q-values based on experienced rewards to guide decision-making in an environment[42].
Deep Q-Networks (DQN)	DQN extend Q-Learning to complex environments using deep neural networks to approximate Q-values. DQNs enable handling high-dimensional state spaces and continuous action spaces, allowing agents to efficiently navigate intricate scenarios by learning from experiences[43].
Policy Gradient Methods	Policy gradient algorithms directly optimize action-selection strategies, aiming to maximize cumulative rewards by computing and updating the policy's parameters through gradient ascent[44].
Model-Based RL	Model-based RL is utilized in reaction optimization, where agents plan chemical reactions to maximize desired outcomes. By simulating various reaction conditions and their potential yields, RL agents identify optimal reaction pathways, expediting the search for efficient chemical transformations[45].
Actor-Critic Methods	Actor-critic combines policy-based and value-based approaches, with the "actor" selecting actions and the "critic" estimating expected rewards. This hybrid framework provides a balance between exploration and stability[46].

analysis, k-nearest neighbors, support vector machines, Bayes generalized linear model, single-layer neural network, and the random forest algorithm. They obtained data by high throughput screening on three 1536-well plates consisting of a full matrix of 15 aryl and heteroaryl halides, 4 Buchwald ligands, 3 bases, and 23 isoxazole additives, a total of 4608 reactions (including controls). When trained with atomic, molecular, and vibrational descriptors, the random forest model showed the best predictive performance with a root mean square error (RMSE) of 7.8% and a coefficient of determination (R2) value of 0.92. They also found that decrease in size of data significantly affects the prediction accuracy.

Yada et al. employed logistic regression analysis, combined with the utilization of least absolute shrinkage and selection operator (LASSO), to optimize models for forecasting reaction yields in the tungsten-catalyzed epoxidation of alkenes using phosphonic acid catalysts (Fig. 4) [72]. A set of 30 parameters, encompassing diverse electronic and vibrational characteristics of the phosphonic acid catalysts, was amassed through density functional theory (DFT) simulations. For highly active

Table 2
Molecular representation in machine learning.

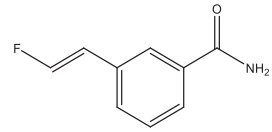
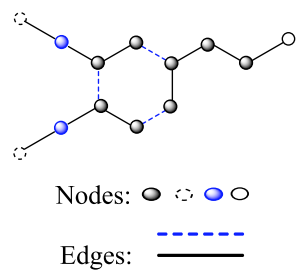
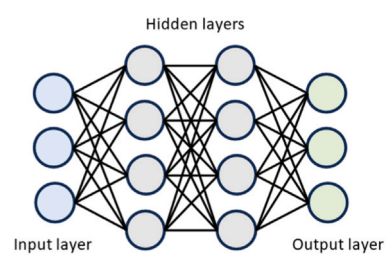
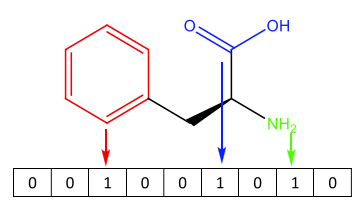
Representation	Description	Example
SMILES and InChI	Introduced in the 1980 s, the simplified molecular-input line-entry system (SMILES) and the international chemical identifier (InChI) were developed as concise and human-readable ways to represent molecular structures. They represent molecules using a compact string of characters that encode atom and bond information[52].	 **SMILES:** NC(C1=CC(/C=C/F)=CC=C1)=O **InChI:** InChI=1S/C10H8FNO2/c11-9-6-8(7-13)4-5-10(9)12/h4-7H,1-3H3/b8-6+
Molecular Graphs	A molecular graph is a graphical representation of a molecule, where each atom is represented as a node, and each bond between atoms is represented as an edge connecting the corresponding nodes. The graph captures the connectivity and structure of the molecule, providing information about the arrangement of atoms and bonds. This representation is widely used in cheminformatics and computational chemistry to describe molecular structures[53].	
Graph Neural Networks	Graph neural networks (GNNs) are a class of machine learning models designed to work with graph-structured data, such as molecular graphs. GNNs are neural network architectures that operate directly on graphs and learn to process and extract information from graph-structured data. GNNs consist of layers that update node representations by aggregating information from neighboring nodes, effectively capturing local and global structural features within the graph. GNNs have gained popularity in tasks involving graph data, including molecule property prediction, drug discovery, and molecular property optimization[54].	
Molecular Fingerprints	Molecular fingerprints represent a molecule as a binary or integer vector, where each element in the vector corresponds to a specific substructure or molecular property. The presence or absence of a particular substructure or property is indicated by the value of the corresponding element in the vector. Molecular fingerprints capture information about the presence of specific functional groups, atom types, and other structural features. They are useful for measuring the similarity between molecules and for quickly identifying molecules with similar substructures. Molecular fingerprints are broader and can include various substructure types and molecular properties, providing a global view of the molecule's structure[55].	

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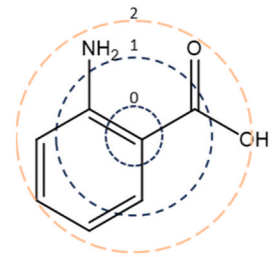
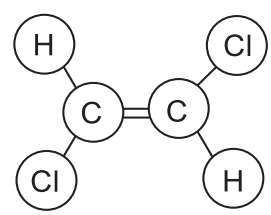
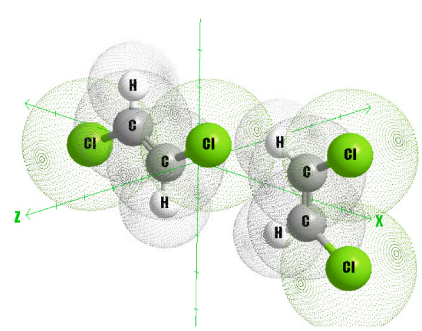
Table 2 (*continued*)

Representation	Description	Example
Circular Fingerprints	Circular fingerprints, also known as circular substructure fingerprints or circular Morgan fingerprints, are a specific type of molecular fingerprint. They are derived from a molecule by considering a set of atom-centered substructures of a specified radius (usually 2 or 3 bonds away from the central atom). These substructures are hashed to create a fixed-length binary or integer vector, where each element corresponds to a hashed substructure. Circular fingerprints capture the presence or absence of specific substructures centered around each atom in the molecule, providing a way to represent the local environment of each atom. Circular fingerprints, on the other hand, focus on local substructures around each atom, providing a more detailed representation of the molecule's local environment[56].	
2D Coordinates	2D coordinates are commonly used to represent molecules as simplified planar graphs. Each atom is a node, and bonds between atoms are edges. This representation captures the connectivity between atoms and is suitable for tasks such as molecular property prediction, similarity analysis, and virtual screening. 2D coordinates are particularly useful when the focus is on capturing molecular topology and identifying common substructures.	
3D Coordinates	3D coordinates provide a more detailed representation of molecular structures, capturing the spatial arrangement of atoms in three dimensions. This representation is crucial for tasks that involve molecular interactions, conformational changes, and docking simulations. In machine learning, 3D coordinates are used to predict protein-ligand binding affinities, analyze molecular interactions, and simulate molecular dynamics.	
Descriptors	Descriptors, in the context of machine learning and chemistry, are numerical values that quantify specific properties or features of molecules. Descriptors provide a compact representation of molecular attributes that can be used as input features for machine learning models. They help to capture various aspects of a molecule's structure and properties in a format that algorithms can process. It is important to know that 2D or 3D coordinates provide spatial information about atom positions, while descriptors offer numerical summaries of molecular properties. The choice between using coordinates or descriptors depends on the specific task at hand: coordinates are important when spatial relationships and angles matter, while descriptors are used to quantify and represent molecular characteristics for analysis and prediction. Each descriptor can be extracted from a molecule using specific calculations or algorithms that analyze the molecular structure and properties. Number of descriptors used in chemoinformatic are in thousands, but their use depends on the type of study and objectives. Here are some common types of descriptors but not limited to:	

0	1	2
C. ar (Sp^2)	C. ar (Sp^2)	C. ar (Sp^2)
	C. ar (Sp^2)	C. ar (Sp^2)
	C. (Sp^2)	N (Sp^3)
		O (Sp^2)
		O (Sp^3)

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Table 2 (*continued*)

Representation	Description	Example
	• Topological descriptors: These capture molecular connectivity information, such as atom and bond counts, molecular weight, and graph-based properties like Wiener index or molecular diameter. • Geometric descriptors: These represent 2D and 3D spatial arrangements of atoms, including bond lengths, angles, dihedral angles, and torsion angles. • Electrostatic descriptors: Descriptors like partial charges, dipole moments, and electrostatic potentials capture the electronic distribution within molecules. • Hydrophobicity descriptors: These describe the hydrophobic or hydrophilic character of molecules, often used in drug design and bioactivity prediction. • Quantum mechanical descriptors: Derived from quantum mechanical calculations, these include properties like molecular orbital energies, electronic densities, and atomic polarizabilities. • Functional group counts: Counting occurrences of specific functional groups in a molecule provides insights into its chemical properties and reactivity. • Molecular connectivity indices: Descriptors like Randic, Balaban, and Zagreb indices quantify molecular structure and connectivity. • Constitutional descriptors: These include various counts and indices related to atoms, bonds, and connectivity within a molecule. • Fingerprint-based descriptors: Molecular fingerprints encode the presence or absence of substructures, providing a binary representation useful for similarity analysis. • 3D pharmacophore descriptors: These encode spatial relationships of chemical features relevant for ligand-receptor interactions. • Atom-centered descriptors: Descriptors calculated for individual atoms, like atomic charges, polarizabilities, and atomic masses. • Drug-likeness descriptors: These capture molecular properties indicative of a molecule's potential as a drug candidate, like Lipinski's Rule of Five descriptors. • Physicochemical descriptors: These encompass a wide range of properties, including solubility, logP (partition coefficient), refractivity, and polar surface area. These descriptors are typically calculated using computational chemistry software, libraries, or programming tools specifically designed for molecular analysis [57,58]. The calculated values serve as input features for machine learning models in various cheminformatics and computational chemistry applications.	

Table 3
Some common tools for extracting molecular descriptors or properties.

Tools	Description
RDKit	A popular open-source cheminformatics toolkit that provides functions to compute molecular descriptors, generate fingerprints, and perform various analyses[59,60].
Open Babel	An open-source chemical toolbox that can calculate diverse molecular descriptors, convert file formats, and perform chemical transformations[61].
ChemDes	A web-based platform that provides a wide range of molecular descriptors, including topological, geometric, and quantum mechanical descriptors[62].
CDK (Chemistry Development Kit)	A Java library with functions for molecular descriptor calculation, substructure search, and various cheminformatics tasks[63].
PaDEL-Descriptor	A software that calculates numerous molecular descriptors, including 1D, 2D, and 3D descriptors, from molecular structures[64].
OEChem (OpenEye Chemistry)	A suite of cheminformatics tools that includes functions for descriptor calculation, molecular visualization, and data analysis[65].
Gaussian	A widely used quantum chemistry software suite that can calculate molecular properties and quantum mechanical descriptors[66].
Molinspiration:	A web-based platform that offers access to a variety of molecular descriptors, drug-likeness calculations, and property predictions[67].
Dragon	Dragon descriptors are numbers that describe a molecule's features. The software calculates these values, helping chemists predict properties and design molecules for tasks like drug development[68].
Mordred	Mordred is a Python tool that computes numbers describing chemical properties of molecules. It's used in drug discovery and chemistry to quickly analyze large sets of molecules for their size, shape, and other features, helping with predictions and research[69].
Cinfony	A Python library, facilitates seamless access to multiple cheminformatics toolkits and resources, including Open Babel, Chemistry Development Kit (CDK), RDKit, Indigo, JChem, OPSIN, and online chemical webservices [70].

phosphonic acids, the root means square error (RMSE) value stood at 26%. However, in the prediction model developed without the inclusion of fluctuation parameters by LASSO, the RMSE increased to 35%. This outcome underscores the necessity of devising appropriate novel parameters for enhancing predictive models.

Schwaller et al. demonstrated that the architectures, commonly used for language tasks, can also be effective in predicting yields [73]. They designed a regression model to predict reaction yields based on reaction SMILES representations. The based their models on reaction fingerprint (rxnfp) model which utilizes a bidirectional encoder representation transformers (BERT) encoder which are usually used in natural language processing. They trained the model on high-throughput experimentation (HTE) data sets for Buchwald-Hartwig and Suzuki-Miyaura reactions, as well as patent data from the USPTO data set. The model showed high prediction performance on first two data sets but was not consistent on USPTO data set because of difference in mass scale in data set.

In a separate study, Doyle group harnessed the power of machine learning to forecast reaction yields, particularly focusing on the intricate realm of Ni-catalyzed Suzuki-Miyaura cross-coupling reactions [74]. The study identified crucial features for modeling phosphine ligand reactivity by exploring remote steric effects and their impact on reaction yields. They used a combination of steric and electronic features to train predictive models, selecting essential features from 13 phosphine ligands like Tolman cone angle, percent buried volume, and minimum electrostatic potential were found to be crucial for explaining the variance in reactivity. Density functional theory (DFT) computations were employed to create informative representations of substrates, catalysts, and reagent, and adding a second steric feature (% Vbur) and an electronic feature (Vmin), which enhanced the model's predictive capability for reaction yields.

6. Selectivity

Selectivity is a critical factor in designing efficient and sustainable chemical processes, as reactions that exhibit high selectivity are more economical, environmentally friendly, and lead to higher-quality products. Achieving high (enantio-, regio-, chem-, etc.) selectivity often involves optimizing reaction conditions, catalysts, and reactant ratios to steer the reaction toward the desired outcome while minimizing competing reactions. ML-driven selectivity predictions can offer efficient resource utilization, reduced waste, and enhanced efficiency. Here we discuss different ML approaches used in selectivity predictions and their efficiencies.

Doyle group developed an open-source Experimental Design via Bayesian Optimization platform (EDBO+) [75] and successfully applied to Ni/photoredox-catalyzed enantioselective cross-electrophile coupling reaction of styrene oxides and aryl iodides in identifying reaction conditions that surpassed the originally reported conditions, resulting in improved yields and enantioselectivities within a 15 and 24 experiments (for two different aryl iodide substrates) among a total of 1728 possible conditions (Fig. 5). The model was trained on data sets of Pd-catalyzed Suzuki-Miyaura coupling and Pd-catalyzed C-H arylation reaction with 352 datapoints, including 11 ligands, 4 solvents, and 8 bases. The study also highlighted the importance of proper initialization methods and

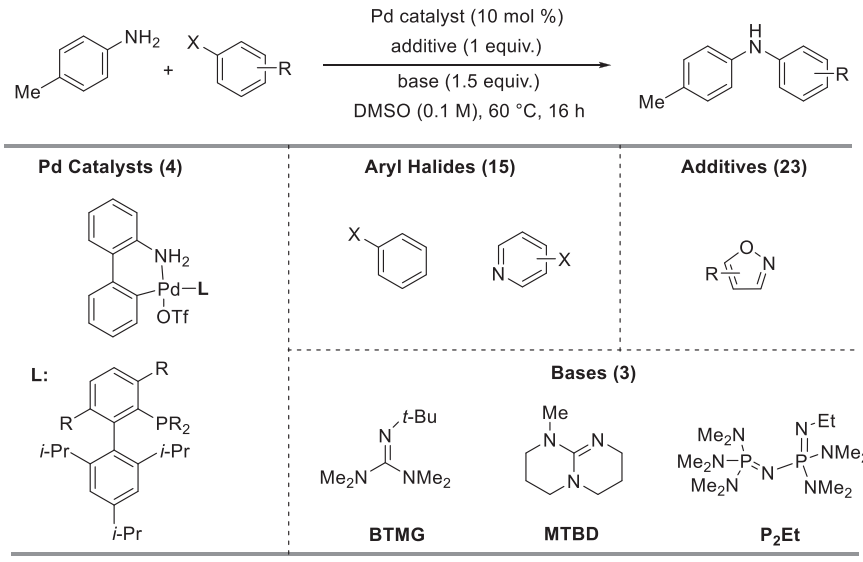


Fig. 3. Reaction components of Buchwald-Hartwig amination for simultaneous evaluation in yield prediction by ML. Adapted from reference [71].

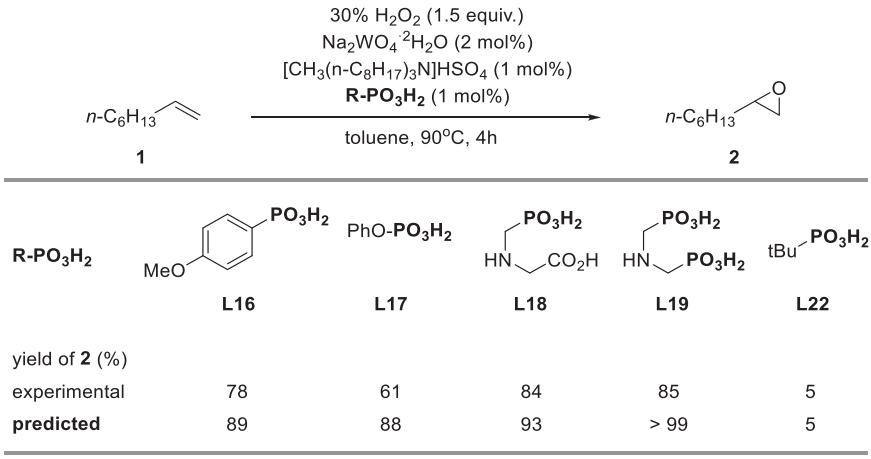
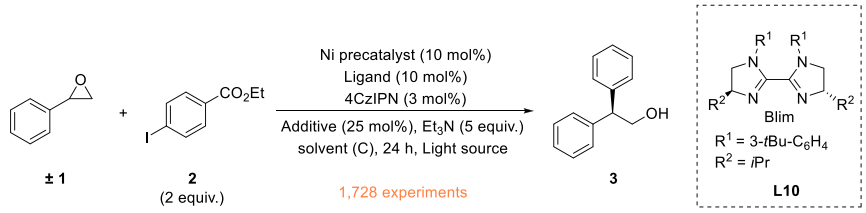


Fig. 4. Comparison of some experimental yields with predicted yields tungsten-catalyzed epoxidation. Adapted from reference [72].

batch sizes in the optimization process. They utilized different feature and featurization methods to represent the reaction components in the data set including one-hot encoding (OHE), quantum mechanics-based features from density functional theory (DFT) calculations, and chemical informatics-based features using Mordred featurization.

Dotson et al. used classification models to determine ligand reactivity and regression models to correlate ligand parameters with regioselectivity in the multi-objective optimization of catalytic reactions using chiral bisphosphine ligands in two sequential reactions [76], the Hayashi-Heck reaction and hydroformylation reaction, and identified improved ligands for both reactions. Ligand parameters were regressed to the experimentally measured $\Delta\Delta G^{\ddagger}$ values using multivariate linear regression (MLR). For the Hayashi-Heck reaction, two parameters, anisotropic phosphorus NMR shielding (aniso31 P NMR) and the occupancy of σ * orbitals of P-C bonds (P − C σ * occupancy), were correlated with the regioselectivity. Principal component analysis (PCA) was used as a mapping technique in the study. The descriptor categories were submitted to PCA, resulting in the generation of a 3D map based on the first principal component of each parameter subset. The resulting designer PCA map was compared to a conventional PCA map that incorporated all descriptors from the data library. Additionally, 2D cross-sections of the 3D space were plotted to further analyze the ligand location in the designer PCA map. The logistic regression classification model used for predicting reactivity and the MLR models for regioselectivity showed promising results while it achieved a 95% prediction accuracy when additional ligands were experimentally evaluated.

Specifically focusing on the S_NAr reactions (nucleophilic aromatic substitution including regioselective nucleophilic aromatic substitution reactions) Guan et al. proposed a workflow consists of three main components: Fast GNN + QM screener, discriminator based on tolerance interval analysis and automated calculator for activation energy barrier [77]. Their Fast GNN + QM screener utilized a graph neural network (GNN) to predict the regioselectivity of nucleophilic aromatic substitution (S_NAr) reactions. It combines machine-learned graph representations with calculated chemical descriptors to predict the relative selectivity scores for different reacting sites. While discriminator based on tolerance interval analysis filters out less confident predictions by implementing a tolerance interval analysis. Predictions with scores below a specified threshold were flagged as less confident and subjected to further analysis. Finally automated calculator for activation energy barrier calculates the activation energy barrier for each reacting site for reactions with less confident predictions. This step involves performing



Ni precatalyst	Ligand	Additive	Solvent	Conc. (M)	Light source	**3** Yield (%)	**3** ee (%)
$NiBr_2$•diglyme	L10	$MgCl_2$	THF	0.050	Photoreactor	70	91
$Ni(cod)_2$	L10	$MgCl_2$	THF	0.033	Photoreactor	80	91

Fig. 5. Applications of EDBO+ : Ni/photoredox-catalyzed enantioselective cross-electrophile coupling of styrene oxides and aryl iodides. Adapted from reference [75].

density functional theory (DFT) calculations to model the reaction pathways and determine the major product. The workflow was tested on both the Pfizer in-house dataset and the USPTO open-access dataset, achieving high prediction accuracies for the major products.

To predict the enantioselectivity in known asymmetric hydrogenation reactions Singh et al. trained a model using random forest (RF) algorithm along with decision tree (DT), extreme gradient boosting (GB), and convolutional neural networks (CNN) [78]. Training with a total of 368 known asymmetric hydrogenation reactions catalyzed by five different axially chiral binaphthyl catalyst families, specifically BINOL-phosphite, BINOL-phosphoramidite, BINAP, BINAPO, and BINOL-phosphoric acid, RF model had the lowest root-mean-square error (RMSE), with an average RMSE of 8.4 ± 1.8. The predicted enantiomeric excess (% ee) values for majority of the samples were within 10 units of the actual % ee values. While DT, GB, and CNN had slightly higher RMSE values ranging from 9.2 ± 1.9 to 11.6 ± 2.8.

Zhang et al. used another data-driven approach to design new chiral carboxylic acids for the construction of indoles with C-central and C-N axial chirality via cobalt catalysis [79]. They used the support vector regression (SVR) model trained on a series of steric and electronic features to describe the influence of the N-substituent of the chiral carboxylic acid (CCA). The ML model accurately predicted the enantioselectivities of different CCAs for the target catalytic transformation. The predictions of the ML model were further corrected using the delta learning method to capture the perturbation of the structure-performance relationship. They conducted virtual screening, involving 360 potential chiral carboxylic acids for the desired atroposelective C–H alkylation of indoles. Subsequently, the three highest-ranked acids projected by the model were synthesized and subjected to thorough experimental assessment, among them, the 3-furyl substituted acid exhibited the 89% enantioselectivity (Fig. 6).

For predicting the face-selectivity of Diels-Alder cycloadditions (DA) and Michael additions (MA) Moskal et al. adopted a combination of machine learning and conformational analysis [80]. The atomic contact vectors (descriptors) representing non-covalent interactions involved in the reactions were used for training the model using various machine learning algorithms such as logistic regression, random forest, multi-layer perceptron, and deep neural network. Here again random forest classifiers performed the best among the machine learning models used. Models which were trained and tested on the DA and MA, achieved prediction accuracies of about 90–91% and 88–89% respectively. However, another model which attempted transfer learning between reaction types, was much less accurate with prediction accuracies of only 55–65% which signifies that the models trained on one reaction type to predict outcomes of the other reaction type (transfer learning) performed poorly, indicating a lack of transferability between different reaction mechanisms.

Quantum or atomistic ML models which map three-dimensional molecular structures depending on the laws of physics offers another complementary strategy for prediction of molecular properties which can help identifying enantioselective organocatalysts. Gallarati used this approach for the development of a stepwise route for accurate prediction of the enantiomeric excess of Lewis base-catalyzed propargylation reactions [81]. The used Spectral London and Axilrod-Teller-Muto (SLATM) molecular representations to identify the intermediates associated with the enantio-determining step. Subsequently, by utilizing the difference between two SLATM representations as input, they derived a set of features that effectively correlate with the accurate prediction of activation energy to be trained on kernel ridge regression (KRR) also known as least squares support vector machines (LS-SVM) algorithm.

Multi-instance learning (MIL) is a machine learning paradigm where the goal is to learn from a set of labeled "bags" or groups of instances rather than individual instances. Varnek et al. applied this approach to predictive modeling of the enantioselectivity of chiral catalysts in asymmetric thiol addition to N-acylimines [82]. For catalyst 3D multi-instance descriptors known as “pmapper” descriptors for representing the ensembles of conformations of catalysts were employed as they captured the physicochemical properties and stereoconfigurations of the catalyst molecules. For the reaction, ISIDA fragment descriptors representing the catalyzed chemical reactions, were used. They found that MIL approach with pmapper descriptors outperformed models based on 3D descriptors calculated with RDKit.

To establish a quantitative link between the structural features of chemical compounds and their selectivity towards specific biological or chemical targets is utmost important in pharmaceutical research. Yu applied support vector regression (SVR) algorithm to develop and improve the prediction performance of quantitative structure-selectivity relationship (QSSR) models for enantioselectivities in the fluorination of allylic alcohols catalyzed by boronic acids and chiral phosphoric acids (Fig. 7) [83]. A SVM model based on three dragon molecular descriptors, achieved determination coefficients (R^2) of 0.951 and 0.945 for the training and test sets respectively, indicating high accuracy in prediction. Another SVM model based on a subset of descriptors including non-covalent interaction, sterimol parameters, and infrared intensity, achieved R^2 values of 0.870 and 0.884 for the training and test sets, respectively. Both SVM models outperformed the MLR models in accurate predictions of enantioselectivity in the fluorination process.

Another QSSR models for predicting enantioselectivities in thiol addition to N-acylimines catalyzed by chiral phosphoric acids has been developed by Yu [84]. The study also used support vector machine (SVM) along with random forest (RF), to build models based on molecular descriptors calculated from the catalysts. With stepwise multiple linear regression (MLR), among 606 dragon molecular descriptors together with 1075 experimental enantioselectivities ($\Delta\Delta G$), 10 structural molecular descriptors were chosen for model trainings. The prediction performance of the QSSR models constructed using SVM and RF algorithms surpasses that of the MLR model, suggesting the existence of nonlinear connections between structural descriptors and the

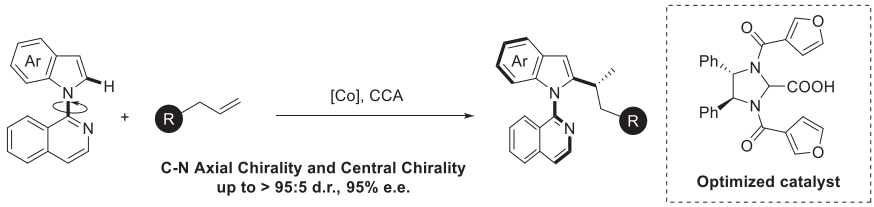


Fig. 6. Data-driven design of new chiral carboxylic for construction of indoles.
Adapted from reference [79].

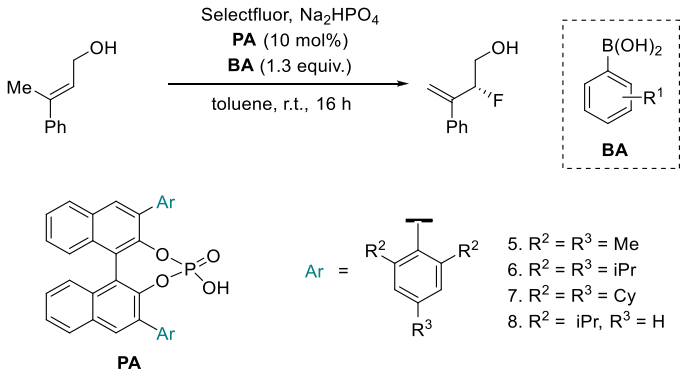


Fig. 7. Reaction scheme and catalysts in fluorination of allylic alcohols.
Adapted from Reference [83].

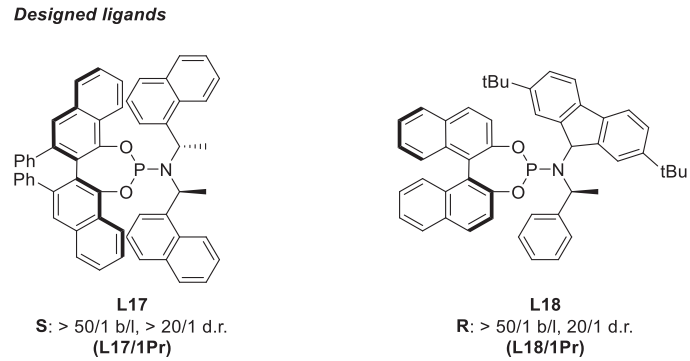


Fig. 8. The optimized ligands for stereodivergent, asymmetric Ir/B hybrid catalysis.
Adapted from reference [85].

enantioselectivities ΔΔG.

To achieve stereodivergent synthesis of α-allyl carboxylic acids by optimizing catalyst structures Chen et al. used regression-based data-driven molecular design approach known as three-dimensional quantitative structure-selectivity relationships (3D-QSSR) [85]. This approach utilizes molecular field analysis (MFA) to analyze the relationship between the reaction outcomes and molecular fields calculated from 3D molecular structures using important structural descriptors which primarily represented steric effects but also capture positional information (3D coordinates) of atoms to describe intermolecular non-covalent interactions. The importance of these descriptors was highlighted in the regression-based optimization of the catalyst structures for achieving improved selectivity in the asymmetric synthesis of α-allyl carboxylic acids using iridium/boron hybrid catalysis with designed ligands, achieving excellent enantioselectivity, with reactions yielding products with enantiomeric excess (ee) of 98% or higher (Fig. 8).

In order to compare the performance of different descriptors in for predicting enantioselectivity in rhodium catalyzed asymmetric allylic substitution (RhCASA) reactions Simon Woodward used linear regression (LR), random forest regression (RF), and gradient boosted regression (GBR to develop models [86]. They used van der Waal volume and Hammett parameter as descriptors to capture steric and electronic effects of ligands, substrates and organo-boron reagent from 610 reactions, and compared their performance with dragon and signature [87] descriptors and found that the formal descriptors provided similar accuracy of prediction while offering more meaningful understanding compared to the later descriptors. Here again RF was found to be more accurate in enantioselectivity prediction as compared to other 2 models for all cyclic and acyclic substrates.

In another work choice of descriptors showed its effects on prediction of enantioselectivity while training the ML models. For enhancing the enantioselectivity of "bis-quat" phase transfer catalyst (PTC) for the intramolecular 1,4-addition which is used to prepare letermovir Merk & Co. trained models with initial 177 catalysts using carhart atom pairs (AP), atom pairs layered with atom types (denoted as DP), and molecular operating environment (MOE) 2D descriptors [88]. With RF and SVM, and CoMFA algorithms a large library was built after successive prediction and screening in 5 rounds. The screening results in each round led to the discovery of new PTCs with higher selectivity, although the improvements were modest. In this study the DP descriptors outperformed other sets of descriptors, including AP (Carhart atom pairs), MOE 2D descriptors, physical-based descriptors, fragment-based descriptors, and ECFP6 fingerprints. While the RF model based on DP descriptors achieved the highest cross-validated R^2 value of 0.49, indicating a good performance in prediction. The study also highlighted that descriptors representing fluoroalkyl content and specific representations of fluorinated substituents, such as CF_3 on R^1 arenes connected to CF_3 on R^2 aromatics, as well as the presence of perfluoroalkyl groups and benzyl groups, accounted for increased product enantioselectivities (Fig. 9). On the other hand, the presence of methoxy groups, carboxy carbons, or nitro groups was shown to drive selectivity lower.

Quantum mechanical calculations provide insights into the energy levels at atomic and subatomic level are much relevant for developing prediction models. In such an effort Das et al. developed a machine learning workflow to predict the enantioselectivity (ee) of relay Heck reactions [89]. In this study, 137 descriptors included physical organic descriptors derived from quantum mechanics (QM) calculations as well as binary vectors obtained through one-hot encoding to represent the presence or absence of reaction components were employed. Highly correlated features with correlation coefficients higher than 0.9 were identified and only one feature from each correlated set was retained for feature reduction which led to 91 features, which were used to train a new random forest (RF) model. Interestingly, the reduction of the feature space resulted in a slightly lower performance with a test RMSE of 8.4 ± 1.5 compared to 8.0 ± 1.3 obtained with the full feature space. It was also observed that the inclusion of 28 additional global reactivity features like molecular electronegativity, global hardness, softness, and electronic spatial extent did not improve the accuracy of predictions.

7. Retrosynthesis

Retrosynthesis, a fundamental approach in the realm of chemistry, has embraced two distinct methodologies: the rule-based and the rule-

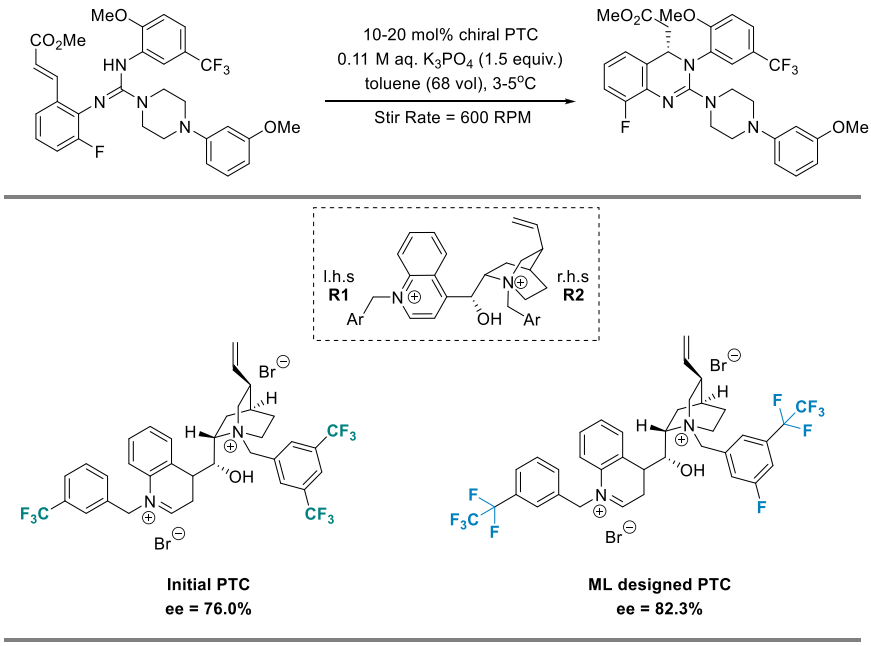


Fig. 9. Comparison of initial best PTC candidate with the best performer from the ML design exercise.
Adapted from reference [88].

free methods, each carrying its unique characteristics and challenges. The rule-based approach mirrors the decision-making process of organic chemists, where predefined reaction patterns are applied to specific synthetic goals. However, the creation of expert-coded rules demands significant time and expertise, acting as a potential bottleneck in the process [90]. In contrast, the rule-free method offers an innovative alternative by directly associating target compounds with potential starting materials [91]. This novel approach treats molecules as linguistic entities, using techniques akin to natural language processing to predict feasible retrosynthetic paths. The utilization of Molecular Transformers, which have exhibited prowess in handling forward reactions and retrosynthetic predictions, has been a significant breakthrough in this direction [92].

Several notable retrosynthetic models have emerged from the molecular transformers' paradigm. The self-correcting retrosynthesis predictor (SCROP), developed by Zheng et al., employs a transformer neural network to predict retrosynthetic pathways [93]. This model translates retrosynthetic planning into a symbolic language problem, achieving impressive accuracy (Fig. 10).

Another advancement is RetroPrime, a template-free, transformer-based method, addressing limitations of output diversity and chemical feasibility [94]. The investigation into the effects of text-like reaction representations (SMILES) and the transformer architecture by Tetko et al. highlighted the role of natural language processing in retrosynthetic predictions [95].

Despite these advancements, challenges persist. While some models have achieved high accuracies, the applicability of proprietary training and test sets raises questions about generalizability. Recent developments explore graph-enhanced transformers and hybrid models, striving for improved diversity in reactant suggestions. Notably, most transformer-based retrosynthesis methods are currently limited to single-step predictions.

A notable contribution is the molecular transformer model's integration, which facilitates automated retrosynthesis planning through hypergraph exploration, reducing human intervention. Additionally, the evolution of single-step models to predict not only reactants but also reagents, solvents, and catalysts represents a significant leap forward in retrosynthesis technology, paving the way for more comprehensive and accurate predictions. As the field continues to evolve, the synergy between linguistic models, advanced neural networks, and chemical

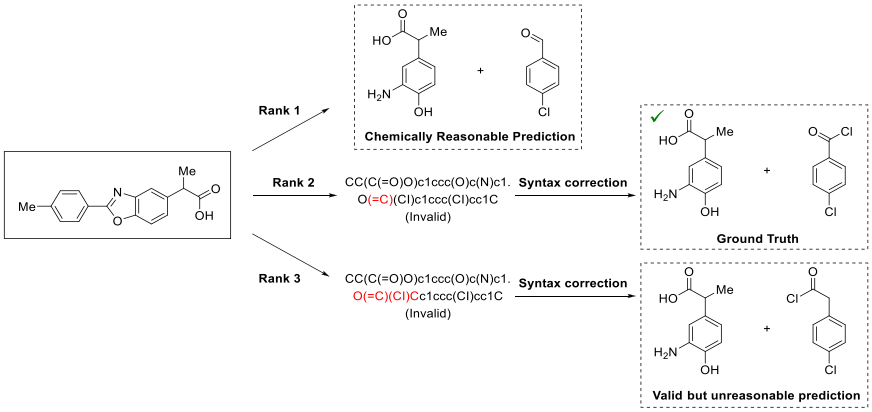


Fig. 10. Retrosynthetic predictions coupling with the syntax corrector, ground truth achieved by syntax correction.
Adapted from reference [93].

insights holds promising potential for reshaping retrosynthesis strategies.

Hiromasa Kaneko applied sequence-to-sequence (Seq2Seq) with attention technique to predict reactants from monomer structures corresponding to polymers, as well as to predict monomer structures from reactants for proposing monomer structures with free bonds for polymer design (Fig. 11). This technique uses recurrent neural networks (RNNs) for language translation tasks. Seq2Seq with attention allows for the input of a sequence of words and the output of a sequence of words [96].

Collaborations Pharmaceuticals developed MegaSyn suite to propose drug analogs, synthetic route, and rank their synthetic feasibility before ultimately selecting molecules to synthesize and testing in vitro. They cured a set of retrosynthetic reactions for retrosynthetic analysis by applying a set of transformations that apply known reactions in reverse. A set of large data set of reactions along with hand-written SMIRKS representing common organic synthesis transformations was reversed to obtain retrosynthetic reactions [97].

In the context of reaction pathway analysis and synthesis accessibility, Li et al. developed a predictive model to address the specific task of identifying the shortest reaction paths (SRP) and quantifying the feasibility of synthesis. These models were meticulously constructed by extracting templates from an extensive chemical reaction network, drawing on diverse data sources including the USPTO (United States Patent and Trademark Office) and the pistachio reaction dataset. The outcome of this endeavor was the establishment of a comprehensive reaction knowledge graph, facilitating a deeper understanding of reaction pathways and synthesis accessibility. This innovation presents a noteworthy contribution to the field's efforts in unraveling efficient chemical transformations. [98]. The graph captures the relationships between reactants and products and provides a structured representation of the chemical reaction network (Fig. 12). The reaction knowledge graph can be used for various purposes, such as identifying starting materials, analyzing reaction pathways, predicting compound synthesis accessibility, and aiding in retrosynthesis analysis.

Integration of quantum chemical calculations for retrosynthesis and synthesis planning was attempted by Toniato et al. in an AI-QC (artificial intelligence-quantum chemistry) framework. They combined data-driven AI models for retrosynthetic analysis with quantum chemical calculations to validate and prioritize suggested reaction steps. The framework could fill the gap of missing data and prediction uncertainty by leveraging first-principles calculations, specifically utilizing the SCINE chemoton platform for reaction explorations. For a single-step reaction, specifically a Williamson ether synthesis to produce ethoxybenzene AI model suggested multiple reaction variants with different bases and solvents and for more complex Friedel-Crafts acylation reaction, where regioselectivity and reactivity considerations were significant factors, architecture struggled to predict the correct regioisomer depending on the directing group attached to the aromatic ring [99].

A convolutional neural network (CNN) based feature extraction network was proposed by Liu et al. for retrosynthesis predictions. This network extracts multi-scale features from molecular descriptors by utilizing filters with different sizes. Additionally, a two-branch feature extraction layer is used to merge the multi-scale features of several molecular descriptors for retrosynthesis prediction. The model was trained on benchmark chemical datasets such as USPTO-50k [100] , and it achieved improved accuracy compared to existing models. The model was tested for bioretrosynthesis prediction using the MetaNetX metabolic and demonstrated superior performance [101].

Leveraging atom environments as descriptors to predict reactant candidates for retrosynthetic analysis were employed by Ucak et al. to build RetroTRAE model. The model was trained using a dataset from the USPTO and achieved a top-1 accuracy of 58.3% on the test dataset. It outperformed other state-of-the-art neural machine translation-based methods (Fig. 13) [102].

8. Drug design

The landscape of drug discovery has been profoundly reshaped by the integration of machine learning methodologies. Within this multidisciplinary realm, machine learning has ushered in a diverse array of applications, revolutionizing the pace and precision of drug development. In this section the key domains where machine learning has proven instrumental in expediting drug discovery efforts are explored briefly.

8.1. *Ligand-based drug design*

Within this approach, machine learning algorithms delve into extensive databases housing known small-molecule ligands and their corresponding biological activities. Through discerning patterns and interconnections within these datasets, machine learning forecasts the activity of fresh compounds and prioritizes potential drug candidates for rigorous investigation [103–105].

8.2. *Structure-based drug design*

The distinctive approach of utilizing three-dimensional structures of target proteins characterizes machine learning in structure-based drug design. In this domain, the focus revolves around predicting binding affinity with potential drug candidates. This involves investigating

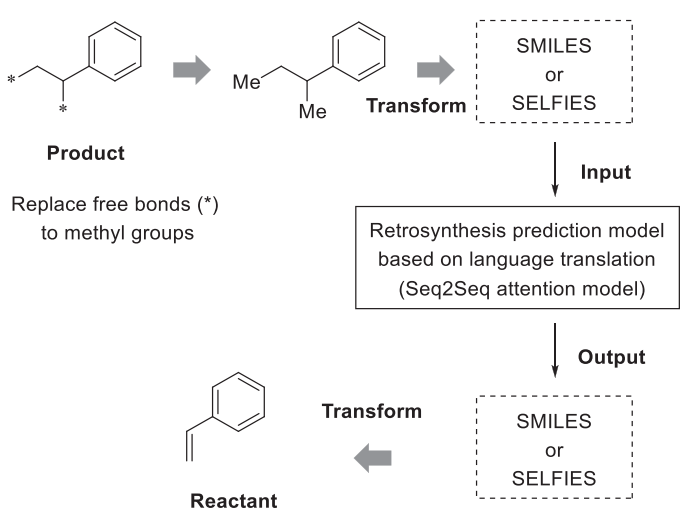


Fig. 11. Flow of reactant prediction using retrosynthesis prediction model for polymer compounds.
Adapted from reference [96].

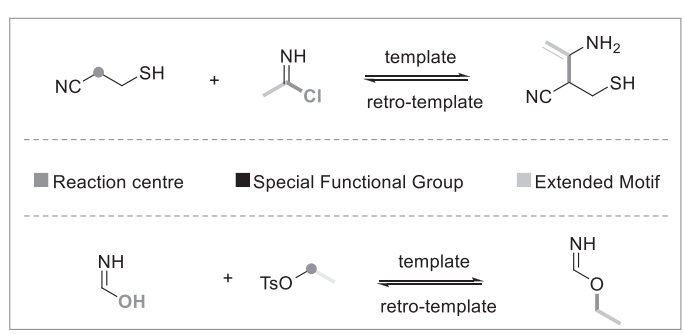


Fig. 12. Examples of template extraction, in which colored sites are added to the reaction template. The red part refers to the reaction centers where the reaction occurs; blue part refers to the special functional groups around the reaction center; black part.
Adapted from reference [98].

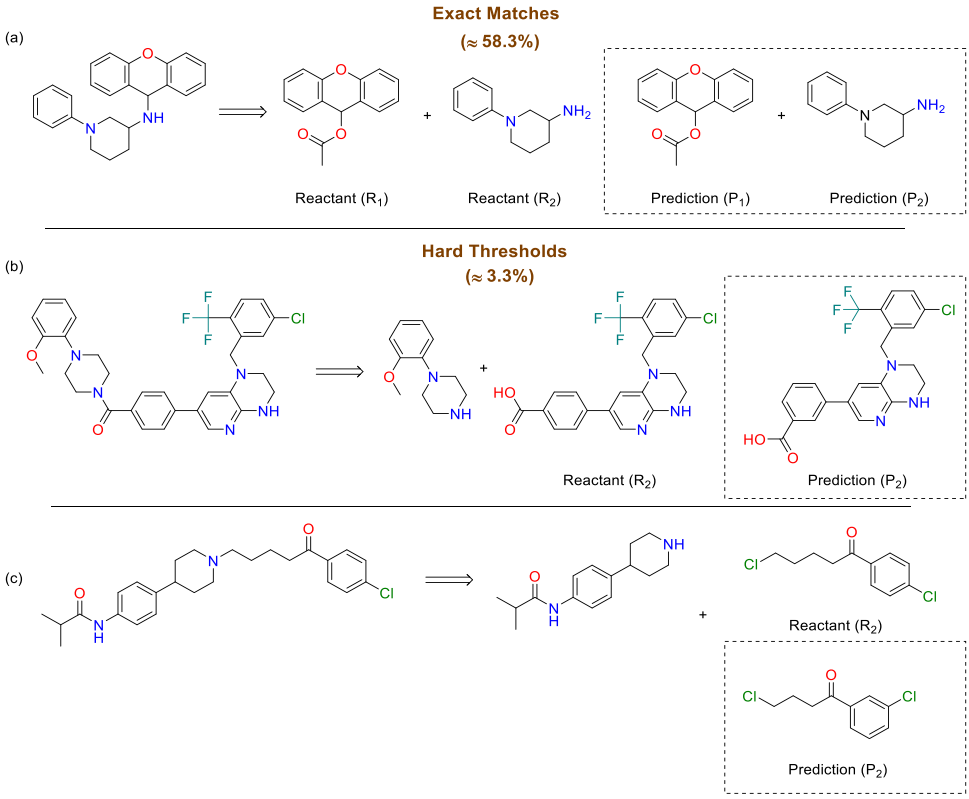


Fig. 13. Representative examples of (a) exact predictions, (b) predictions with a single, and (c) double fragment mutations. Reproduced from reference [102] (CC 3.0).

protein-ligand interactions, pinpointing binding sites, and orchestrating molecular docking simulations to meticulously optimize drug design [106–108].

8.3. *Virtual screening*

Venturing into virtual screening, machine learning algorithms navigate sprawling chemical libraries, deftly identifying potential drug candidates characterized by heightened affinity and selectivity for target proteins. Through this avenue, the computational realm drastically reduces the set of compounds necessitating experimental scrutiny, thereby conserving invaluable time and resources [109,110].

8.4. *De novo drug design*

Embarking on de novo drug design, machine learning techniques craft novel molecules endowed with specific properties. By sculpting the chemical space and implementing generative algorithms, this avenue tailors molecular structures based on distinct criteria. These criteria span target affinity, pharmacokinetic properties, and safety profiles [111,112].

8.5. *Binding site detection*

These binding sites play a crucial role in various aspects of drug discovery and molecular interaction studies. The goal of binding site detection is to predict these sites accurately and efficiently, providing valuable insights into molecular recognition and facilitating the design of targeted therapeutics. Based on the molecular representations these approaches rely on.

- Sequence-based models: These models predict binding sites by analyzing the sequence of amino acids in the protein. They can use techniques like image segmentation or deep learning architectures such as convolutional neural networks (CNNs) to identify residues involved in binding sites [113–115].
- Surface-based models: These models represent the protein structure as continuous shapes characterized by geometric and physicochemical features. They can use methods like geodesic CNNs or DeepSurf to determine interaction fingerprints or discretize the protein surface to predict binding sites. These approaches capture the structural and functional relationships between atoms and their interactions on the protein surface [116–118].
- Structure-based models: These models utilize the 3D structure of the protein to detect binding sites. Methods like interatomic gap volumes or regions of buried pocket surfaces are used to identify potential binding pockets. Deep learning approaches can also be applied to learn directly from the protein's raw structure, using techniques like graph neural networks (GNNs) to capture the relationships within the protein structure and discover cryptic binding pockets [119–121].

8.6. *Drug-target interaction prediction*

Drug-target interaction (DTI) prediction refers to the process of predicting the interactions between drug molecules and target proteins. It involves determining the binding affinity or bioactivity of a given set of molecules on one or more macromolecular targets.

- Sequence-based approaches: These models use amino acid sequences along with additional ligand representations to make predictions. They often employ 1D convolutional neural networks (CNNs) or recurrent neural networks (RNNs) to learn from the string representations of both ligands and proteins [113,122–124].
- Structure-based approaches: These approaches leverage the 3D structure of proteins to predict DTIs. They can use voxelized protein

representations combined with CNNs to capture spatial dependencies and local patterns important for binding affinity prediction and pocket detection [125–127].

- Graph-based approaches: Graph neural networks (GNNs) are employed in these approaches to operate on molecular graphs and capture the structural and functional relationships between atoms in molecular complexes. GNNs are versatile and suitable for various tasks in structure-based drug discovery [128].

8.7. *Protein-based de novo molecule design*

Protein-based de novo molecule design refers to the generation of novel chemical entities with desired properties from scratch, using proteins as the starting point. It involves leveraging the information from protein structures and their interactions with ligands to design molecules. There are several protein-based de novo molecule design approaches.

- Sequence-based approaches: These methods treat the problem as a machine translation task, where the amino acid sequences of proteins and SMILES strings of ligands are considered as "sentences" in different languages to be matched. Transformer architectures and reinforcement learning techniques have been used in this approach [129,130].
- 3D structure-based approaches: These approaches utilize the tertiary structure information of proteins to generate molecules in the form of 3D ligands or strings. Deep learning models, such as convolutional neural networks (CNNs) and graph neural networks, have been used to predict binding sites and discover cryptic binding pockets [131–133].

9. Material design

Machine learning drives transformative progress across diverse domains like photovoltaics, OLEDs, high-energy materials, and polymer design. It optimizes solar systems, revolutionizes OLED development, transforms materials research, and accelerates polymer innovation. This integration has fueled the innovation across renewable energy, materials science, and beyond.

9.1. *Organic photovoltaics (OPVs)*

Machine learning is making substantial contributions to the field of photovoltaics, transforming how solar energy systems are designed and operated. By leveraging data-driven insights, machine learning techniques enhance various aspects of photovoltaic technology. Applications of ML in OPVs include predicting solar irradiance levels for optimized energy production [134], analyzing module degradation rates [135], enhancing material stability[136], to construct quantitative structure-property relationship (QSPR) models for OPVs [137], efficiency prediction and optimization of fullerene or nonfullerene materials [138,139], for optimize experimental conditions (such as donor-acceptor ratio, solvent additives, crystallinity, light type, etc.) [140], to predict high-performance formulations [141], and to predict photostability of OPV devices via high-throughput robot [142]. Overall, machine learning's impact in organic photovoltaics has improved efficiency, cost-effectiveness, and the adoption of renewable energy sources.

9.2. *Organic light-emitting diodes (OLEDs)*

Machine learning revolutionizes the development of OLEDs by profoundly impacting their design, efficiency, and overall performance. Through data-driven insights and predictive modeling, machine learning techniques play a central role in optimizing various facets of OLED technology. These encompass the discovery of novel OLED materials by predicting their properties and characteristics [143], the creation of extensive virtual material libraries [144] using quantum simulations and cheminformatics [145], the establishment of relationships between molecular structures and OLED performance metrics [146], and the selection of optimal materials for different OLED components [147]. Furthermore, machine learning predicts device performance metrics, guides design optimization for efficient energy conversion and emission [148], ensures manufacturing quality by identifying defects [149], and accelerates the development of new OLED emitters with specific color emission and efficiency [148,150]. Overall, the integration of machine learning empowers OLED development, leading to enhanced efficiency, innovative design strategies, and the realization of OLED's potential across diverse applications.

9.3. *High energetic materials*

Machine learning is transforming the landscape of high-energy materials research through its versatile applications. By predicting properties [151], establishing molecular relationships [152,153], optimizing synthesis [154], enhancing safety assessment [152], and accelerating material screening [155], machine learning expedites the discovery [156] and development of advanced energetic compounds [157]. It empowers quantum chemistry simulations [158], integrates diverse data sources for knowledge extraction, and aids in designing safer materials with improved properties [159]. Additionally, machine learning facilitates rapid material characterization [160], predicts reaction pathways [161], and contributes to a deeper understanding of material behavior.

9.4. *Polymer design*

Machine learning is also becoming increasingly integral to polymer design, facilitating the rapid development of polymers with specific properties and applications. It achieves this through predicting polymer properties from their chemical structures [162], establishing intricate relationships between polymer structures and properties [163,164], and accelerating the discovery [165] of new polymers by analyzing extensive chemical databases [166]. Machine learning also optimizes polymerization processes [167], enabling the identification of optimal conditions for desired properties [168] while minimizing resource consumption. Integrating diverse data sources like chemical databases and spectroscopic information, machine learning offers a comprehensive understanding of polymer behavior [169]. It also automates high-throughput screening of polymers, swiftly identifying promising candidates for further exploration [170,171]. Furthermore, machine learning guides the formulation of polymer blends and composites, predicting resultant properties based on constituent components [172]. Additionally, high-throughput synthesis has also been showed to merge the computational techniques, robotics, and automation in microwave-assisted Suzuki–Miyaura polycondensation to facilitate the evaluation of over 170 co-polymer photocatalysts [173]. The resulting large data was combined with machine-learning techniques to predict the hydrogen evolution rates of polymer photocatalysts.

Besides polymers, high-throughput systems in combination of ML have also been used for building a library of small molecule semiconductors and optimizing their optical-electronic properties [174]. Hence, machine learning has been demonstrated to significantly expedite material design and advancing materials science across various sectors and industries.

10. Large Language models (LLM)

LLMs are powerful tools in AI that can generate text and answer complex questions. LLMs are built on machine learning to recognize, guess or predict words and create phrases, paragraphs, and full essays that reflect how humans write and speak. Besides their application in

various fields, they have also been applied in chemistry. The applications of LLMs have been used for molecular editing and molecular property prediction but their major application in chemistry is for robotic automation of chemistry experiments.

Liu et al. used a multi-modal approach to develop a model called MoleculeSTM to demonstrate that the LLMs can help in the challenging task of modifying a part of the molecule to get the desired property while keeping the other properties unchanged [175]. Edwards et al. proposed a LLM, MolT5, which can generate description for a given molecule, and can also generate a molecule to match a given description [176].

Hocky et al. [177] and White et al. [178] while working with Codex [179], a GPT-3 based LLM, to generate code for chemistry tasks, successfully generated input files for Gaussian software. By providing a text prompt describing the desired properties of the molecule, the LLM can generate modified versions of the input molecule that meet the desired properties while preserving other desirable characteristics.

Nascimento and Pimentel [180] provided a valuable insight into the understanding of chemistry by LLMs, using the ChatGPT model developed by OpenAI. They find when the ChatGPT-3 was attempted to be used to convert compound names to SMILES representations and vice versa, their accuracy was relatively low. In predicting Octanol–Water partition coefficients RME was 31%, while when GPT-3 model was attempted to get structural information of coordination compounds, 58% hit rate was observed.

Abstraction of synthetic data and conversion into low-level instructions is an important step for robotic automation. In an early adaptation, Cronin group demonstrated that how natural language processing can be used for abstraction of synthetic plans from literature by a robotic laboratory platform for the synthesis of organic compounds, guided by standardized method descriptions using the chemical descriptive language (XDL) [181,182]. A-Lab, an autonomous laboratory for synthesis of inorganic powders by Szymanski et al. also applied natural-language models trained on the literature and were optimized using an active ML approach [183]. In another effort, Yoshikawa et al. demonstrated that a natural language processing system called Clarify can take a chemistry experiment description in natural language as input for robotic tasks [184]. Clarify converts instructions into a structured experiment plan written in XDL, which is a robot programming language. The refined XDL plan is then passed to the task and motion planning (TAMP) module, which generates the low-level robot action plan for execution of chemistry experiment by a robot.

11. Challenges

Although applications of AI and ML is being rapidly adapted by chemistry community but realizing the full potential of AI in chemistry necessitates addressing some key challenges as following.

11.1. *Data availability and Quality*

The availability and quality of data hold paramount importance in building AI models. Reliable predictions and optimizations rely on robust datasets, emphasizing the significance of meticulous data curation and sharing. The challenge lies in obtaining high-quality chemical and biological data, given the inherent variations in experimental conditions and measurement accuracy. Performance of ML models relay on large datasets, but in synthetic chemistry datasets for a particular reaction are usually far less than the optimum data required for near-accurate predictions. Combining data from different sources and published works could greatly affect the reliability of the combined data.

11.2. *Data Representation*

Even after collecting a reliable data, applying machine learning to chemistry involves preprocessing of data, presenting several difficulties. Besides, transforming diverse data formats into a uniform format, selection of effective molecular representation for model training is a challenging task. Choosing the right molecular representation is not a one-size-fits-all endeavor; rather, it requires a nuanced approach that aligns with the objectives of the model and the nuances of the chemical data at hand. Different algorithms may benefit from distinct molecular descriptors or representations, adding another layer of complexity to the selection process. Furthermore, the challenge is exacerbated by the evolving nature of chemistry, where new compound types continually emerge, necessitating adaptability in representation selection.

11.3. *Reproducibility of AI models*

Even if a machine learning algorithm is successful for particular process but if data sources are different reproducibility of models can be affected. Replicating the exact computational environment, including hardware and software configurations, is crucial. Changes in these dependencies may lead to variations in results, especially when utilizing specialized hardware for training. Beside that differences in software library versions and updates can impact the reproducibility of AI models. Ensuring consistent versioning of software and libraries is essential for reproducing results over time. Many machine learning algorithms used in chemistry are sensitive to hyperparameter settings. Small changes in hyperparameters can lead to variations in model performance, making it challenging to reproduce results consistently. Algorithm complexity such as in deep learning models, often have numerous parameters, making them more susceptible to variations in training conditions. This complexity introduces challenges in reproducing the exact training process.

11.4. *Interpretability of AI models*

Another crucial hurdle lies in the interpretability of AI models. Many AI algorithms, including those used in chemistry, often function as black-box systems, obscuring the rationale behind their decisions. This opacity raises concerns about trust and safety and underscores the need to establish transparent and interpretable AI methodologies. Even if a model can quickly and successfully learn and predict the outcome but understanding the mechanism behind the given outcome will remain a big challenge for human understanding.

12. Conclusion

In the realm of organic synthesis, artificial intelligence (AI), especially machine learning (ML), has revolutionized the landscape by offering predictive capabilities for reaction yields, selectivity, retrosynthetic pathways, and even material and drug design. Despite the challenges faced in each of these ML directions within chemistry, the opportunities they present for groundbreaking discoveries and inventions are substantial. As we navigate the evolving intersection of AI and organic synthesis, it becomes clear that addressing these challenges opens the door to innovation. **T**he synergy of AI and organic synthesis promises a future where computational methods play a pivotal role in accelerating the discovery and optimization of chemical processes. Beyond the current hurdles, fostering a collaborative environment that encourages interdisciplinary efforts, promotes transparent reporting of methodologies and results, and supports the sharing of both data and code will be instrumental. Moreover, ensuring broad accessibility to computational resources, especially for researchers with diverse backgrounds and resource constraints, is paramount. With these concerted efforts, the transformative potential of AI in organic synthesis can be fully realized, ushering in an era of unprecedented efficiency, precision, and creativity in the field.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] C.W. Coley, D.A. Thomas, J.A.M. Lummiss, J.N. Jaworski, C.P. Breen, V. Schultz, T. Hart, J.S. Fishman, L. Rogers, H. Gao, et al., A robotic platform for flow synthesis of organic compounds informed by AI planning, Science 365 (2019) eaax1566, https://doi.org/10.1126/science.aax1566.
[2] R.L. Greenaway, K.E. Jelfs, A.C. Spivey, S.N. Yaliraki, From alchemist to AI chemist, Nat. Rev. Chem. 7 (2023) 527–528, https://doi.org/10.1038/s41570-023-00522-w.
[3] L.M. Roch, F. Häse, C. Kreisbeck, T. Tamayo-Mendoza, L.P.E. Yunker, J.E. Hein, A. Aspuru-Guzik, ChemOS: Orchestrating autonomous experimentation, Sci. Robot. 3 (2018) eaat5559, https://doi.org/10.1126/scirobotics.aat5559.
[4] H. Zhu, Big data and artificial intelligence modeling for drug discovery, Insel, P. A. Ed.; Annual Review of Pharmacology and Toxicology, Vol. 60, Annu. Rev. Pharmacol. Toxicol. Vol 60 (2020) 573–589.
[5] S.V. Ley, D.E. Fitzpatrick, R.M. Myers, C. Battilocchio, R.J. Ingham, Machine-assisted organic synthesis, Angew. Chem. Int. Ed. 54 (2015) 10122–10136, https://doi.org/10.1002/anie.201501618.
[6] J. Jiménez-Luna, F. Grisoni, G. Schneider, Drug discovery with explainable artificial intelligence, Nat. Mach. Intell. 2 (2020) 573–584, https://doi.org/10.1038/s42256-020-00236-4.
[7] E. Gawehn, J.A. Hiss, G. Schneider, Deep learning in drug discovery, Mol. Inform. 35 (2016) 3–14, https://doi.org/10.1002/minf.201501008.
[8] X. Yang, Y. Wang, R. Byrne, G. Schneider, S. Yang, Concepts of artificial intelligence for computer-assisted drug discovery, Chem. Rev. 119 (2019) 10520–10594, https://doi.org/10.1021/acs.chemrev.8b00728.
[9] P. Fantke, C. Cinquemani, P. Yaseneva, J. De Mello, H. Schwabe, B. Ebeling, A. A. Lapkin, Transition to sustainable chemistry through digitalization, Chem 7 (2021) 2866–2882, https://doi.org/10.1016/j.chempr.2021.09.012.
[10] R.G. Bergman, R.L. Danheiser, Reproducibility in chemical research, Angew. Chem. Int. Ed. 55 (2016) 12548–12549, https://doi.org/10.1002/anie.201606591.
[11] S.G. Baird, T.D. Sparks, What is a minimal working example for a self-driving laboratory? Matter 5 (2022) 4170–4178, https://doi.org/10.1016/j.matt.2022.11.007.
[12] Y. Shen, J.E. Borowski, M.A. Hardy, R. Sarpong, A.G. Doyle, T. Cernak, Automation and computer-assisted planning for chemical synthesis, Nat. Rev. Methods Prim. 1 (2021) 23, https://doi.org/10.1038/s43586-021-00022-5.
[13] S.V. Ley, The engineering of chemical synthesis: humans and machines working in harmony, Angew. Chem. Int. Ed. 57 (2018) 5182–5183, https://doi.org/10.1002/anie.201802383.
[14] V. Duros, J. Grizou, W. Xuan, Z. Hosni, D.L. Long, H.N. Miras, L. Cronin, Human versus robots in the discovery and crystallization of gigantic polyoxometalates, Angew. Chem. Int. Ed. 56 (2017) 10815–10820, https://doi.org/10.1002/anie.201705721.
[15] M.H.S. Segler, M. Preuss, M.P. Waller, Planning chemical syntheses with deep neural networks and symbolic AI, Nature 555 (2018) 604–610, https://doi.org/10.1038/nature25978.
[16] J.N. Wei, D. Duvenaud, A. Aspuru-Guzik, Neural networks for the prediction of organic chemistry reactions, ACS Cent. Sci. 2 (2016) 725–732, https://doi.org/10.1021/acscentsci.6b00219.
[17] J.E. Hein, Machine learning made easy for optimizing chemical reactions, Nature 590 (2021) 40–41, https://doi.org/10.1038/d41586-021-00209-6.
[18] P.S. Gromski, A.B. Henson, J.M. Granda, L. Cronin, How to explore chemical space using algorithms and automation, Nat. Rev. Chem. 3 (2019) 119–128, https://doi.org/10.1038/s41570-018-0066-y.
[19] H. Gao, T.J. Struble, C.W. Coley, Y. Wang, W.H. Green, K.F. Jensen, Using machine learning to predict suitable conditions for organic reactions, ACS Cent. Sci. 4 (2018) 1465–1476, https://doi.org/10.1021/acscentsci.8b00357.
[20] C.J. Taylor, K.C. Felton, D. Wigh, M.I. Jeraal, R. Grainger, G. Chessari, C. N. Johnson, A.A. Lapkin, Accelerated chemical reaction optimization using multi-task learning, ACS Cent. Sci. 9 (2023) 957–968, https://doi.org/10.1021/acscentsci.3c00050.
[21] M. Jutel, M. Zemelka-Wiacek, M. Ordak, O. Pfaar, T. Eiwegger, M. Rechenmacher, C.A. Akdis, The artificial intelligence (AI) revolution: How important for scientific work and its reliable sharing, Allergy 78 (2023) 2085–2088, https://doi.org/10.1111/all.15778.
[22] G.A. Vouros, Explainable deep reinforcement learning: state of the art and challenges, ACM Comput. Surv. 55 (2022) 1–39, https://doi.org/10.1145/3527448.
[23] F. Oviedo, J.L. Ferres, T. Buonassisi, K.T. Butler, Interpretable and explainable machine learning for materials science and chemistry, Acc. Mater. Res. 3 (2022) 597–607, https://doi.org/10.1021/accountsmr.1c00244.
[24] A.F. de Almeida, R. Moreira, T. Rodrigues, Synthetic organic chemistry driven by artificial intelligence, Nat. Rev. Chem. 3 (2019) 589–604, https://doi.org/10.1038/s41570-019-0124-0.
[25] H.J. Kulik, M.S. Sigman, Advancing discovery in chemistry with artificial intelligence: from reaction outcomes to new materials and catalysts, Acc. Chem. Res. 54 (2021) 2335–2336, https://doi.org/10.1021/acs.accounts.1c00232.
[26] J. Westermayr, M. Gastegger, K.T. Schütt, R.J. Maurer, Perspective on integrating machine learning into computational chemistry and materials science, J. Chem. Phys. 154 (2021), https://doi.org/10.1063/5.0047760.
[27] S. Gow, M. Niranjan, S. Kanza, J.G. Frey, A review of reinforcement learning in chemistry, Digit. Discov. 1 (2022) 551–567, https://doi.org/10.1039/D2DD00047D.
[28] S. Russell, P. Norvig, Artificial Intelligence: A Modern Approach, Global Edition, Pearson Education, 2021.
[29] C. Cortes, V. Vapnik, Support-vector networks, Mach. Learn. 20 (1995) 273–297, https://doi.org/10.1007/BF00994018.
[30] G. Cano, J. Garcia-Rodriguez, A. Garcia-Garcia, H. Perez-Sanchez, J. A. Benediktsson, A. Thapa, A. Barr, Automatic selection of molecular descriptors using random forest: Application to drug discovery, Expert Syst. Appl. 72 (2017) 151–159, https://doi.org/10.1016/j.eswa.2016.12.008.
[31] M. Shen, Y. Xiao, A. Golbraikh, V.K. Gombar, A. Tropsha, Development and validation of k-nearest-neighbor QSPR models of metabolic stability of drug candidates, J. Med. Chem. 46 (2003) 3013–3020, https://doi.org/10.1021/jm020491t.
[32] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, Nature 521 (2015) 436–444, https://doi.org/10.1038/nature14539.
[33] K.P. Sinaga, M.-S. Yang, Unsupervised K-means clustering algorithm, IEEE Access 8 (2020) 80716–80727, https://doi.org/10.1109/ACCESS.2020.2988796.
[34] S. Eschrich, J. Ke, L.O. Hall, D.B. Goldgof, Fast accurate fuzzy clustering through data reduction, IEEE Trans. Fuzzy Syst. 11 (2003) 262–270, https://doi.org/10.1109/TFUZZ.2003.809902.
[35] J. Xu, A. Hagler, Chemoinformatics and drug discovery, Molecules 7 (2002) 566–600, https://doi.org/10.3390/70800566.
[36] I.T. Jolliffe, J. Cadima, Principal component analysis: a review and recent developments, Philos. Trans. A Math. Phys. Eng. Sci. 374 (2016) 20150202, https://doi.org/10.1098/rsta.2015.0202.
[37] A. Giuliani, The application of principal component analysis to drug discovery and biomedical data, Drug Discov. Today 22 (2017) 1069–1076, https://doi.org/10.1016/j.drudis.2017.01.005.
[38] D. Polykovskiy, A. Zhebrak, D. Vetrov, Y. Ivanenkov, V. Aladinskiy, P. Mamoshina, M. Bozdaganyan, A. Aliper, A. Zhavoronkov, A. Kadurin, Entangled conditional adversarial autoencoder for de novo drug discovery, Mol. Pharm. 15 (2018) 4398–4405, https://doi.org/10.1021/acs.molpharmaceut.8b00839.
[39] L. Zhang, J. Tan, D. Han, H. Zhu, From machine learning to deep learning: progress in machine intelligence for rational drug discovery, Drug Discov. Today 22 (2017) 1680–1685, https://doi.org/10.1016/j.drudis.2017.08.010.
[40] Y. Bian, X.Q. Xie, Generative chemistry: drug discovery with deep learning generative models, J. Mol. Model. 27 (2021) 71, https://doi.org/10.1007/s00894-021-04674-8.
[41] P. Schneider, Y. Tanrikulu, G. Schneider, Self-organizing maps in drug discovery: compound library design, scaffold-hopping, repurposing, Curr. Med. Chem. 16 (2009) 258–266, https://doi.org/10.2174/092986709787002655.
[42] Y. Liu, B. Zhang, J. Zhao, W. Wang, Z. Lv, Molecular design based on Q-learning and maximum likelihoodestimation, in: 39th Chinese Control Conference (CCC), 2020, IEEE, 2020, pp. 2119–2124.
[43] B. Tang, J. Ewalt, H.-L. Ng, Generative AI models for drug discovery. Biophysical and Computational Tools in Drug Discovery, Springer, 2021, pp. 221–243.
[44] J. You, B. Liu, Z. Ying, V. Pande, J. Leskovec, Graph convolutional policy network for goal-directed molecular graph generation, Adv. Neural Inf. Process. Syst. 31 (2018).
[45] J.W. Kim, B.J. Park, T.H. Oh, J.M. Lee, Model-based reinforcement learning and predictive control for two-stage optimal control of fed-batch bioreactor, Comput. Chem. Eng. 154 (2021) 107465, https://doi.org/10.1016/j.compchemeng.2021.107465.
[46] J. Elsborg, A. Bhowmik, Equivariant graph-representation-based actor-critic reinforcement learning for nanoparticle design, J. Chem. Inf. Model. 63 (2023) 3731–3741, https://doi.org/10.1021/acs.jcim.3c00394.
[47] R. Asahara, T. Miyao, Extended connectivity fingerprints as a chemical reaction representation for enantioselective organophosphorus-catalyzed asymmetric reaction prediction, ACS Omega 7 (2022) 26952–26964, https://doi.org/10.1021/acsomega.2c03812.
[48] J.A. Beltran, L. Aguilera-Mendoza, C.A. Brizuela, Optimal selection of molecular descriptors for antimicrobial peptides classification: an evolutionary feature weighting approach, BMC Genom. 19 (2018) 672, https://doi.org/10.1186/s12864-018-5030-1.
[49] J.S. Delaney, ESOL: estimating aqueous solubility directly from molecular structure, J. Chem. Inf. Comput. Sci. 44 (2004) 1000–1005, https://doi.org/10.1021/ci034243x.
[50] A. Tayyebi, A.S. Alshami, Z. Rabiei, X. Yu, N. Ismail, M.J. Talukder, J. Power, Prediction of organic compound aqueous solubility using machine learning: a comparison study of descriptor-based and fingerprints-based models, J. Chemin.-. 15 (2023) 99, https://doi.org/10.1186/s13321-023-00752-6.

[51] M.C. Sorkun, A. Khetan, S. Er, AqSolDB, a curated reference set of aqueous solubility and 2D descriptors for a diverse set of compounds, Sci. Data 6 (2019) 143, https://doi.org/10.1038/s41597-019-0151-1.
[52] D. Weininger, SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules, J. Chem. Inf. Comput. Sci. 28 (1988) 31–36, https://doi.org/10.1021/ci00057a005.
[53] S. Kearnes, K. McCloskey, M. Berndl, V. Pande, P. Riley, Molecular graph convolutions: moving beyond fingerprints, J. Comput. -Aided Mol. Des. 30 (2016) 595–608, https://doi.org/10.1007/s10822-016-9938-8.
[54] P. Reiser, M. Neubert, A. Eberhard, L. Torresi, C. Zhou, C. Shao, H. Metni, C. van Hoesel, H. Schopmans, T. Sommer, et al., Graph neural networks for materials science and chemistry, Commun. Mater. 3 (2022) 93, https://doi.org/10.1038/s43246-022-00315-6.
[55] A. Cereto-Massague, M.J. Ojeda, C. Valls, M. Mulero, S. Garcia-Vallve, G. Pujadas, Molecular fingerprint similarity search in virtual screening, Methods 71 (2015) 58–63, https://doi.org/10.1016/j.ymeth.2014.08.005.
[56] R.C. Glem, A. Bender, C.H. Arnby, L. Carlsson, S. Boyer, J. Smith, Circular fingerprints: flexible molecular descriptors with applications from physical chemistry to ADME, IDrugs 9 (2006) 199–204.
[57] A.E. Comesana, T.T. Huntington, C.D. Scown, K.E. Niemeyer, V.H. Rapp, A systematic method for selecting molecular descriptors as features when training models for predicting physiochemical properties, Fuel 321 (2022), https://doi.org/10.1016/j.fuel.2022.123836.
[58] R. Todeschini, V. Consonni, Molecular Descriptors for Chemoinformatics, Wiley, 2009 https://doi.org/DOI: https://doi.org/10.1002/9783527628766.
[59] RDKit: Open-Source Cheminformatics Software. https://www.rdkit.org/ , 2006 (Accessed 24 August 2023).
[60] J.-P. Ebejer, G.M. Morris, C.M. Deane, Freely available conformer generation methods: how good are they? J. Chem. Inf. Model. 52 (2012) 1146–1158, https://doi.org/10.1021/ci2004658.
[61] N.M. O'Boyle, M. Banck, C.A. James, C. Morley, T. Vandermeersch, G. R. Hutchison, Open babel: an open chemical toolbox, J. Chemin.-. 3 (2011) 33, https://doi.org/10.1186/1758-2946-3-33.
[62] J. Dong, D.-S. Cao, H.-Y. Miao, S. Liu, B.-C. Deng, Y.-H. Yun, N.-N. Wang, A.-P. Lu, W.-B. Zeng, A.F. Chen, ChemDes: an integrated web-based platform for molecular descriptor and fingerprint computation, J. Chemin.-. 7 (2015) 60, https://doi.org/10.1186/s13321-015-0109-z.
[63] E.L. Willighagen, J.W. Mayfield, J. Alvarsson, A. Berg, L. Carlsson, N. Jeliazkova, S. Kuhn, T. Pluskal, M. Rojas-Chertó, O. Spjuth, et al., The chemistry development kit (CDK) v2.0: atom typing, depiction, molecular formulas, and substructure searching, J. Chemin.-. 9 (2017) 33, https://doi.org/10.1186/s13321-017-0220-4.
[64] C.W. Yap, PaDEL-descriptor: an open source software to calculate molecular descriptors and fingerprints, J. Comput. Chem. 32 (2011) 1466–1474, https://doi.org/10.1002/jcc.21707.
[65] OEChem (OpenEye Chemistry) , Cheminformatics & Modeling Toolkits. http://www.eyesopen.com , 2023 (Accessed 20 August 2023).
[66] J.B. Foresman, A. Frisch, Exploring chemistry with electronic structure methods, Gaussian Inc,, 2015.
[67] Molinspiration Cheminformatics free web services. https://www.molinspiration.com, , 2023 (Accessed 24 August 2023).
[68] A. Mauri, V. Consonni, M. Pavan, R. Todeschini, Dragon software: an easy approach to molecular descriptor calculations, Match-Commun. Math. Comput. Chem. 56 (2006) 237–248.
[69] H. Moriwaki, Y.-S. Tian, N. Kawashita, T. Takagi, Mordred: a molecular descriptor calculator, J. Chemin.-. 10 (2018) 4, https://doi.org/10.1186/s13321-018-0258-y.
[70] N.M. O'Boyle, G.R. Hutchison, Cinfony - combining Open Source cheminformatics toolkits behind a common interface, Chem. Cent. J. 2 (2008), https://doi.org/10.1186/1752-153x-2-24.
[71] D.T. Ahneman, J.G. Estrada, S. Lin, S.D. Dreher, A.G. Doyle, Predicting reaction performance in C–N cross-coupling using machine learning, Science 360 (2018) 186–190, https://doi.org/10.1126/science.aar5169.
[72] A. Yada, K. Nagata, Y. Ando, T. Matsumura, S. Ichinoseki, K. Sato, Machine learning approach for prediction of reaction yield with simulated catalyst parameters, Chem. Lett. 47 (2018) 284–287, https://doi.org/10.1246/cl.171130.
[73] P. Schwaller, A.C. Vaucher, T. Laino, J.L. Reymond, Prediction of chemical reaction yields using deep learning, Mach. Learn. -Sci. Technol. 2 (2021) 015016, https://doi.org/10.1088/2632-2153/abc81d.
[74] A.M. Zuranski, J.I. Martinez Alvarado, B.J. Shields, A.G. Doyle, Predicting reaction yields via supervised learning, Acc. Chem. Res. 54 (2021) 1856–1865, https://doi.org/10.1021/acs.accounts.0c00770.
[75] J.A.G. Torres, S.H. Lau, P. Anchuri, J.M. Stevens, J.E. Tabora, J. Li, A. Borovika, R.P. Adams, A.G. Doyle, A multi-objective active learning platform and web app for reaction optimization, J. Am. Chem. Soc. 144 (2022) 19999–20007, https://doi.org/10.1021/jacs.2c08592.
[76] J.J. Dotson, L. van Dijk, J.C. Timmerman, S. Grosslight, R.C. Walroth, F. Gosselin, K. Puntener, K.A. Mack, M.S. Sigman, Data-driven multi-objective optimization tactics for catalytic asymmetric reactions using bisphosphine ligands, J. Am. Chem. Soc. 145 (2023) 110–121, https://doi.org/10.1021/jacs.2c08513.
[77] Y. Guan, T. Lee, K. Wang, S. Yu, J.C. McWilliams, SNAr regioselectivity predictions: machine learning triggering DFT reaction modeling through statistical threshold, J. Chem. Inf. Model. 63 (2023) 3751–3760, https://doi.org/10.1021/acs.jcim.3c00580.
[78] S. Singh, M. Pareek, A. Changotra, S. Banerjee, B. Bhaskararao, P. Balamurugan, R.B. Sunoj, A unified machine-learning protocol for asymmetric catalysis as a proof of concept demonstration using asymmetric hydrogenation, Proc. Natl. Acad. Sci. U. S. A. 117 (2020) 1339–1345, https://doi.org/10.1073/pnas.1916392117.
[79] Z.J. Zhang, S.W. Li, J.C.A. Oliveira, Y. Li, X. Chen, S.Q. Zhang, L.C. Xu, T. Rogge, X. Hong, L. Ackermann, Data-driven design of new chiral carboxylic acid for construction of indoles with C-central and C-N axial chirality via cobalt catalysis, Nat. Commun. 14 (2023) 3149, https://doi.org/10.1038/s41467-023-38872-0.
[80] M. Moskal, W. Beker, S. Szymkuc, B.A. Grzybowski, Scaffold-directed face selectivity machine-learned from vectors of non-covalent interactions, Angew. Chem. Int. Ed. 60 (2021) 15230–15235, https://doi.org/10.1002/anie.202101986.
[81] S. Gallarati, R. Fabregat, R. Laplaza, S. Bhattacharjee, M.D. Wodrich, C. Corminboeuf, Reaction-based machine learning representations for predicting the enantioselectivity of organocatalysts, Chem. Sci. 12 (2021) 6879–6889, https://doi.org/10.1039/d1sc00482d.
[82] A. Varnek, D. Zankov, P. Polishchuk, T. Madzhidov, Multi-Instance learning approach to predictive modeling of catalysts enantioselectivity, Synlett 32 (2021) 1833–1836, https://doi.org/10.1055/a-1553-0427.
[83] X. Yu, Support vector machine-based prediction of enantioselectivity in fluorination of allylic alcohols, ChemistrySelect 7 (2022), https://doi.org/10.1002/slct.202104369.
[84] X. Yu, Prediction of enantioselectivity in thiol addition to imines catalyzed by chiral phosphoric acids, J. Phys. Org. Chem. 35 (2022), https://doi.org/10.1002/poc.4338.
[85] H. Chen, S. Yamaguchi, Y. Morita, H. Nakao, X. Zhai, Y. Shimizu, H. Mitsunuma, M. Kanai, Data-driven catalyst optimization for stereodivergent asymmetric synthesis by iridium/boron hybrid catalysis, Cell Rep. Phys. Sci. 2 (2021), https://doi.org/10.1016/j.xcrp.2021.100679.
[86] B. Owen, K. Wheelhouse, G. Figueredo, E. Özcan, S. Woodward, Machine learnt patterns in rhodium-catalysed asymmetric Michael addition using chiral diene ligands, Results Chem. 4 (2022), https://doi.org/10.1016/j.rechem.2022.100379.
[87] P. Carbonell, L. Carlsson, J.-L. Faulon, Stereo signature molecular descriptor, J. Chem. Inf. Model. 53 (2013) 887–897, https://doi.org/10.1021/ci300584r.
[88] K.W. Lexa, K.M. Belyk, J. Henle, B. Xiang, R.P. Sheridan, S.E. Denmark, R. T. Ruck, E.C. Sherer, Application of machine learning and reaction optimization for the iterative improvement of enantioselectivity of cinchona-derived phase transfer catalysts, Org. Process Res. Dev. 26 (2021) 670–682, https://doi.org/10.1021/acs.oprd.1c00155.
[89] M. Das, P. Sharma, R.B. Sunoj, Machine learning studies on asymmetric relay Heck reaction-Potential avenues for reaction development, J. Chem. Phys. 156 (2022), https://doi.org/10.1063/5.0084432.
[90] T. Klucznik, B. Mikulak-Klucznik, M.P. McCormack, H. Lima, S. Szymkuc, M. Bhowmick, K. Molga, Y. Zhou, L. Rickershauser, E.P. Gajewska, et al., Efficient syntheses of diverse, medicinally relevant targets planned by computer and executed in the laboratory, Chem 4 (2018) 522–532, https://doi.org/10.1016/j.chempr.2018.02.002.
[91] C.D. Christ, M. Zentgraf, J.M. Kriegl, Mining electronic laboratory notebooks: analysis, retrosynthesis, and reaction based enumeration, J. Chem. Inf. Model. 52 (2012) 1745–1756, https://doi.org/10.1021/ci300116p.
[92] K. Lin, Y. Xu, J. Pei, L. Lai, Automatic retrosynthetic route planning using template-free models, Chem. Sci. 11 (2020) 3355–3364, https://doi.org/10.1039/c9sc03666k.
[93] S. Zheng, J. Rao, Z. Zhang, J. Xu, Y. Yang, Predicting retrosynthetic reactions using self-corrected transformer neural networks, J. Chem. Inf. Model. 60 (2020) 47–55, https://doi.org/10.1021/acs.jcim.9b00949.
[94] X. Wang, Y. Li, J. Qiu, G. Chen, H. Liu, B. Liao, C.-Y. Hsieh, X. Yao, RetroPrime: a diverse, plausible and transformer-based method for single-step retrosynthesis predictions, Chem. Eng. J. 420 (2021), https://doi.org/10.1016/j.cej.2021.129845.
[95] I.V. Tetko, P. Karpov, R. Van Deursen, G. Godin, State-of-the-art augmented NLP transformer models for direct and single-step retrosynthesis, Nat. Commun. 11 (2020) 5575, https://doi.org/10.1038/s41467-020-19266-y.
[96] H. Taniwaki, H. Kaneko, Retrosynthetic and synthetic reaction prediction model based on sequence-to-sequence with attention for polymer designs, Macromol. Theory Simul. 32 (2023), https://doi.org/10.1002/mats.202300011.
[97] F. Urbina, C.T. Lowden, J.C. Culberson, S. Ekins, MegaSyn: integrating generative molecular design, automated analog designer, and synthetic viability prediction, ACS Omega 7 (2022) 18699–18713, https://doi.org/10.1021/acsomega.2c01404.
[98] B. Li, H. Chen, Prediction of compound synthesis accessibility based on reaction knowledge graph, Molecules 27 (2022), https://doi.org/10.3390/molecules27031039.
[99] A. Toniato, J.P. Unsleber, A.C. Vaucher, T. Weymuth, D. Probst, T. Laino, M. Reiher, Quantum chemical data generation as fill-in for reliability enhancement of machine-learning reaction and retrosynthesis planning, Digit Discov. 2 (2023) 663–673, https://doi.org/10.1039/d3dd00006k.
[100] B. Liu, B. Ramsundar, P. Kawthekar, J. Shi, J. Gomes, Q. Luu Nguyen, S. Ho, J. Sloane, P. Wender, V. Pande, Retrosynthetic reaction prediction using neural sequence-to-sequence models, ACS Cent. Sci. 3 (2017) 1103–1113, https://doi.org/10.1021/acscentsci.7b00303.
[101] F. Yang, J. Liu, Q. Zhang, Z. Yang, X. Zhang, CNN-based two-branch multi-scale feature extraction network for retrosynthesis prediction, BMC Bioinforma. 23 (2022) 362, https://doi.org/10.1186/s12859-022-04904-7.

[102] U.V. Ucak, I. Ashyrmamatov, J. Ko, J. Lee, Retrosynthetic reaction pathway prediction through neural machine translation of atomic environments, Nat. Commun. 13 (2022), https://doi.org/10.1038/s41467-022-28857-w.
[103] D. Merk, L. Friedrich, F. Grisoni, G. Schneider, De Novo design of bioactive small molecules by artificial intelligence, Mol. Inform. 37 (2018), https://doi.org/10.1002/minf.201700153.
[104] A. Zhavoronkov, Y.A. Ivanenkov, A. Aliper, M.S. Veselov, V.A. Aladinskiy, A. V. Aladinskaya, V.A. Terentiev, D.A. Polykovskiy, M.D. Kuznetsov, A. Asadulaev, et al., Deep learning enables rapid identification of potent DDR1 kinase inhibitors, Nat. Biotechnol. 37 (2019) 1038, https://doi.org/10.1038/s41587-019-0224-x.
[105] W. Yuan, D. Jiang, D.K. Nambiar, L.P. Liew, M.P. Hay, J. Bloomstein, P. Lu, B. Turner, Q.T. Le, R. Tibshirani, et al., Chemical space mimicry for drug discovery, J. Chem. Inf. Model. 57 (2017) 875–882, https://doi.org/10.1021/acs.jcim.6b00754.
[106] L. Kurgan, F.M. Disfani, Structural protein descriptors in 1-dimension and their sequence-based predictions, Curr. Protein Pept. Sci. 12 (2011) 470–489, https://doi.org/10.2174/138920311796957711.
[107] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Zidek, A. Potapenko, et al., Highly accurate protein structure prediction with AlphaFold, Nature 596 (2021) 583–589, https://doi.org/10.1038/s41586-021-03819-2.
[108] M. Baek, F. DiMaio, I. Anishchenko, J. Dauparas, S. Ovchinnikov, G.R. Lee, J. Wang, Q. Cong, L.N. Kinch, R.D. Schaeffer, et al., Accurate prediction of protein structures and interactions using a three-track neural network, Science 373 (2021) 871–876, https://doi.org/10.1126/science.abj8754.
[109] M. Stahl, M. Rarey, Detailed analysis of scoring functions for virtual screening, J. Med. Chem. 44 (2001) 1035–1042, https://doi.org/10.1021/jm0003992.
[110] F. Sohraby, H. Aryapour, Rational drug repurposing for cancer by inclusion of the unbiased molecular dynamics simulation in the structure-based virtual screening approach: challenges and breakthroughs, Semin. Cancer Biol. 68 (2021) 249–257, https://doi.org/10.1016/j.semcancer.2020.04.007.
[111] M.H.S. Segler, T. Kogej, C. Tyrchan, M.P. Waller, Generating focused molecule libraries for drug discovery with recurrent neural networks, ACS Cent. Sci. 4 (2018) 120–131, https://doi.org/10.1021/acscentsci.7b00512.
[112] M. Olivecrona, T. Blaschke, O. Engkvist, H.M. Chen, Molecular de-novo design through deep reinforcement learning, J. Chemin.-. 9 (2017), https://doi.org/10.1186/s13321-017-0235-x.
[113] M. Karimi, D. Wu, Z.Y. Wang, Y. Shen, DeepAffinity: interpretable deep learning of compound-protein affinity through unified recurrent and convolutional neural networks, Bioinformatics 35 (2019) 3329–3338, https://doi.org/10.1093/bioinformatics/btz111.
[114] M. Tsubaki, K. Tomii, J. Sese, Compound-protein interaction prediction with end-to-end learning of neural networks for graphs and sequences, Bioinformatics 35 (2019) 309–318, https://doi.org/10.1093/bioinformatics/bty535.
[115] V. Gligorijevic, P.D. Renfrew, T. Kosciolek, J.K. Leman, D. Berenberg, T. Vatanen, C. Chandler, B.C. Taylor, I.M. Fisk, H. Vlamakis, et al., Structure-based protein function prediction using graph convolutional networks, Nat. Commun. 12 (2021), https://doi.org/10.1038/s41467-021-23303-9.
[116] P. Gainza, F. Sverrisson, F. Monti, E. Rodola, D. Boscaini, M.M. Bronstein, B. E. Correia, Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning, Nat. Methods 17 (2020) 184, https://doi.org/10.1038/s41592-019-0666-6.
[117] S.K. Mylonas, A. Axenopoulos, P. Daras, DeepSurf: a surface-based deep learning approach for the prediction of ligand binding sites on proteins, Bioinformatics 37 (2021) 1681–1690, https://doi.org/10.1093/bioinformatics/btab009.
[118] F. Sverrisson, J. Feydy, B.E. Correia, M.M. Bronstein, S.O.C. Ieee Comp, Fast end-to-end learning on protein surfaces, Jun 19-25, in: In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021, Electr Network, 2021, pp. 15267–15276. Jun 19-25.
[119] J. Jimenez, S. Doerr, G. Martinez-Rosell, A.S. Rose, G. De Fabritiis, DeepSite: protein-binding site predictor using 3D-convolutional neural networks, Bioinformatics 33 (2017) 3036–3042, https://doi.org/10.1093/bioinformatics/btx350.
[120] A. Meller, M.D. Ward, J.H. Borowsky, J.M. Lotthammer, M. Kshirsagar, F. Oviedo, J.L. Ferres, G. Bowman, Predicting the locations of cryptic pockets from single protein structures using the PocketMiner graph neural network, Biophys. J. 122 (2023) 445A, https://doi.org/10.1016/j.bpj.2022.11.2400.
[121] I. Kozlovskii, P. Popov, Protein-peptide binding site detection using 3D convolutional neural networks, J. Chem. Inf. Model. 61 (2021) 3814–3823, https://doi.org/10.1021/acs.jcim.1c00475.
[122] H. Ozturk, A. Ozgur, E. Ozkirimli, DeepDTA: deep drug-target binding affinity prediction, Bioinformatics 34 (2018) 821–829, https://doi.org/10.1093/bioinformatics/bty593.
[123] R. Ozcelik, H. Ozturk, A. Ozgur, E. Ozkirimli, ChemBoost: a chemical language based approach for protein - ligand binding affinity prediction, Mol. Inform. 40 (2021), https://doi.org/10.1002/minf.202000212.
[124] I. Lee, J. Keum, H. Nam, DeepConv-DTI: prediction of drug-target interactions via deep learning with convolution on protein sequences, Plos Comput. Biol. 15 (2019), https://doi.org/10.1371/journal.pcbi.1007129.
[125] J. Lim, S. Ryu, K. Park, Y.J. Choe, J. Ham, W.Y. Kim, Predicting drug-target interaction using a novel graph neural network with 3D structure-embedded graph representation, J. Chem. Inf. Model. 59 (2019) 3981–3988, https://doi.org/10.1021/acs.jcim.9b00387.
[126] H. Cho, S.J. Hyeon, J.Y. Shin, V.E. Alvarez, T.D. Stein, J. Lee, N.W. Kowall, A. C. McKee, H. Ryu, J.S. Seo, Alterations of transcriptome signatures in head trauma-related neurodegenerative disorders, Sci. Rep. 10 (2020), https://doi.org/10.1038/s41598-020-65916-y.
[127] W. Torng, R.B. Altman, Graph convolutional neural networks for predicting drug-target interactions, J. Chem. Inf. Model. 59 (2019) 4131–4149, https://doi.org/10.1021/acs.jcim.9b00628.
[128] Q.Q. Liu, P.S. Wang, C.J. Zhu, B.B. Gaines, T. Zhu, J.B. Bi, M.H. Song, OctSurf: efficient hierarchical voxel-based molecular surface representation for protein-ligand affinity prediction, J. Mol. Graph. Model. 105 (2021), https://doi.org/10.1016/j.jmgm.2021.107865.
[129] D. Grechishnikova, Transformer neural network for protein-specific de novo drug generation as a machine translation problem, Sci. Rep. 11 (2021), https://doi.org/10.1038/s41598-020-79682-4.
[130] G. Uludogan, E. Ozkirimli, K.O. Ulgen, N. Karal, A. Ozgur, Exploiting pretrained biochemical language models for targeted drug design, Bioinformatics 38 (2022) ii155–ii161, https://doi.org/10.1093/bioinformatics/btac482.
[131] M. Skalic, A. Varela-Rial, J. Jimenez, G. Martinez-Rosell, G. De Fabritiis, LigVoxel: inpainting binding pockets using 3D-convolutional neural networks, Bioinformatics 35 (2019) 243–250, https://doi.org/10.1093/bioinformatics/bty583.
[132] M. Ragoza, T. Masuda, D.R. Koes, Generating 3D molecules conditional on receptor binding sites with deep generative models, Chem. Sci. 13 (2022) 2701–2713, https://doi.org/10.1039/d1sc05976a.
[133] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial networks, Commun. Acm 63 (2020) 139–144, https://doi.org/10.1145/3422622.
[134] J. Munshi, W. Chen, T. Chien, G. Balasubramanian, Transfer learned designer polymers for organic solar cells, J. Chem. Inf. Model. 61 (2021) 134–142, https://doi.org/10.1021/acs.jcim.0c01157.
[135] T.W. David, G.A. Soares, N. Bristow, D. Bagnis, J. Kettle, Predicting diurnal outdoor performance and degradation of organic photovoltaics via machine learning; relating degradation to outdoor stress conditions, Prog. Photovolt.: Res. Appl. 29 (2021) 1274–1284, https://doi.org/10.1002/pip.3453.
[136] T.W. David, H. Anizelli, T.J. Jacobsson, C. Gray, W. Teahan, J. Kettle, Enhancing the stability of organic photovoltaics through machine learning, Nano Energy 78 (2020) 105342, https://doi.org/10.1016/j.nanoen.2020.105342.
[137] J. Feng, H. Wang, Y. Ji, Y. Li, Molecular design and performance improvement in organic solar cells guided by high-throughput screening and machine learning, Nano Sel. 2 (2021) 1629–1641, https://doi.org/10.1002/nano.202100006.
[138] M.-H. Lee, Robust random forest based non-fullerene organic solar cells efficiency prediction, Org. Electron. 76 (2020) 105465, https://doi.org/10.1016/j.orgel.2019.105465.
[139] M.-Y. Sui, Z.-R. Yang, Y. Geng, G.-Y. Sun, L. Hu, Z.-M. Su, Nonfullerene acceptors for organic photovoltaics: from conformation effect to power conversion efficiencies prediction, Sol. RRL 3 (2019) 1900258, https://doi.org/10.1002/solr.201900258.
[140] X. Rodríguez-Martínez, E. Pascual-San-José, M. Campoy-Quiles, Accelerating organic solar cell material's discovery: high-throughput screening and big data, Energy Environ. Sci. 14 (2021) 3301–3322, https://doi.org/10.1039/D1EE00559F.
[141] N.G. An, J.Y. Kim, D. Vak, Machine learning-assisted development of organic photovoltaics via high-throughput in situ formulation, Energy Environ. Sci. 14 (2021) 3438–3446, https://doi.org/10.1039/D1EE00641J.
[142] X. Du, L. Lüer, T. Heumueller, J. Wagner, C. Berger, T. Osterrieder, J. Wortmann, S. Langner, U. Vongsaysy, M. Bertrand, et al., Elucidating the full potential of opv materials utilizing a high-throughput robot-based platform and machine learning, Joule 5 (2021) 495–506, https://doi.org/10.1016/j.joule.2020.12.013.
[143] Y. Zhao, C. Fu, L. Fu, Y. Liu, Z. Lu, X. Pu, Data-driven machine learning models for quick prediction of thermal stability properties of OLED materials, Mater. Today Chem. 22 (2021) 100625, https://doi.org/10.1016/j.mtchem.2021.100625.
[144] A. Saeki, K. Kranthiraja, A high throughput molecular screening for organic electronics via machine learning: present status and perspective, Jpn. J. Appl. Phys. 59 (SD) (2020) 0801, https://doi.org/10.7567/1347-4065/ab4f39.
[145] R. Gomez-Bombarelli, J. Aguilera-Iparraguirre, T.D. Hirzel, D. Duvenaud, D. Maclaurin, M.A. Blood-Forsythe, H.S. Chae, M. Einzinger, D.G. Ha, T. Wu, et al., Design of efficient molecular organic light-emitting diodes by a high-throughput virtual screening and experimental approach, Nat. Mater. 15 (2016) 1120, https://doi.org/10.1038/nmat4717.
[146] H. Abroshan, P. Winget, H.S. Kwak, Y. An, C.T. Brown, M.D. Halls, Machine Learning for the Design of Novel Oled Materials. in Machine Learning in Materials Informatics: Methods and Applications, in: ACS Symposium Series, Vol. 1416, American Chemical Society, 2022, pp. 33–49.
[147] Y. Zhao, C. Fu, L. Fu, Z. Lu, X. Pu. Machine Learning based Framework for Quick Prediction of Tg and Td of OLED Materials. (2021). https://doi.org/10.26434/chemrxiv-2021-j5pfd.
[148] M.H. Lee, Identification of host-guest systems in green TADF-based OLEDs with energy level matching based on a machine-learning study, Phys. Chem. Chem. Phys. 22 (2020) 16378–16386, https://doi.org/10.1039/d0cp02871a.
[149] V.H. Nguyen, V.H. Pham, X. Cui, M. Ma, H. Kim, Design and evaluation of features and classifiers for OLED panel defect recognition in machine vision, J. Inf. Telecommun. 1 (2017) 334–350, https://doi.org/10.1080/24751839.2017.1355717.
[150] M.A.Bin Janai, K.L. Woon, C.S. Chan, Design of efficient blue phosphorescent bottom emitting light emitting diodes by machine learning approach, Org. Electron. 63 (2018) 257–266, https://doi.org/10.1016/j.orgel.2018.09.029.

[151] D.C. Elton, Z. Boukouvalas, M.S. Butrico, M.D. Fuge, P.W. Chung, Applying machine learning techniques to predict the properties of energetic materials, Sci. Rep. 8 (2018) 9059, https://doi.org/10.1038/s41598-018-27344-x.
[152] Q. Deng, J. Hu, L. Wang, Y. Liu, Y. Guo, T. Xu, X. Pu, Probing impact of molecular structure on bulk modulus and impact sensitivity of energetic materials by machine learning methods, Chemom. Intell. Lab. Syst. 215 (2021) 104331, https://doi.org/10.1016/j.chemolab.2021.104331.
[153] X.-l Tian, S.-w Song, F. Chen, X.-j Qi, Y. Wang, Q.-h Zhang, Machine learning-guided property prediction of energetic materials: Recent advances, challenges, and perspectives, Energ. Mater. Front. (2022), https://doi.org/10.1016/j.enmf.2022.07.005.
[154] X. Zang, X. Zhou, H. Bian, W. Jin, X. Pan, J. Jiang, M.Y. Koroleva, R. Shen, Prediction and construction of energetic materials based on machine learning methods, Molecules 28 (2022) 322, https://doi.org/10.3390/molecules28010322.
[155] Y. Liu, O.C. Esan, Z. Pan, L. An, Machine learning for advanced energy materials, Energy AI 3 (2021) 100049, https://doi.org/10.1016/j.egyai.2021.100049.
[156] D. Walters, N. Rai, O. Sen, W. Lee Perry, Toward a machine-guided approach to energetic material discovery, J. Appl. Phys. 131 (2022), https://doi.org/10.1063/5.0084973.
[157] C. Chen, Y. Zuo, W. Ye, X. Li, Z. Deng, S.P. Ong, A critical review of machine learning of energy materials, Adv. Energy Mater. 10 (2020) 1903242, https://doi.org/10.1002/aenm.201903242.
[158] D.C. Elton, Z. Boukouvalas, M.S. Butrico, M.D. Fuge, P.W. Chung, Applying machine learning techniques to predict the properties of energetic materials, Sci. Rep. 8 (2018) 9059, https://doi.org/10.1038/s41598-018-27344-x.
[159] P.C.H. Nguyen, Y.-T. Nguyen, P.K. Seshadri, J.B. Choi, H.S. Udaykumar, S. Baek, A physics-aware deep learning model for energy localization in multiscale shock-to-detonation simulations of heterogeneous energetic materials, Propellants, Explos., Pyrotech. 48 (2023) e202200268, https://doi.org/10.1002/prep.202200268.
[160] X.-l Tian, S.-w Song, F. Chen, X.-j Qi, Y. Wang, Q.-h Zhang, Machine learning-guided property prediction of energetic materials: Recent advances, challenges, and perspectives, Energ. Mater. Front. 3 (2022) 177–186, https://doi.org/10.1016/j.enmf.2022.07.005.
[161] L. Cao, J. Zeng, B. Wang, T. Zhu, J.Z.H. Zhang, Ab initio neural network MD simulation of thermal decomposition of a high energy material CL-20/TNT, Phys. Chem. Chem. Phys. 24 (2022) 11801–11811, https://doi.org/10.1039/D2CP00710J.
[162] W. Sha, Y. Li, S. Tang, J. Tian, Y. Zhao, Y. Guo, W. Zhang, X. Zhang, S. Lu, Y.-C. Cao, et al., Machine learning in polymer informatics, InfoMat 3 (2021) 353–361, https://doi.org/10.1002/inf2.12167.
[163] K. Ishikiriyama, Polymer informatics based on the quantitative structure-property relationship using a machine-learning framework for the physical properties of polymers in the ATHAS data bank, Thermochim. Acta 708 (2022) 179135, https://doi.org/10.1016/j.tca.2021.179135.
[164] R. Ma, Z. Liu, Q. Zhang, Z. Liu, T. Luo, Evaluating polymer representations via quantifying structure–property relationships, J. Chem. Inf. Model. 59 (2019) 3110–3119, https://doi.org/10.1021/acs.jcim.9b00358.
[165] P. Xu, H. Chen, M. Li, W. Lu, New opportunity: machine learning for polymer materials design and discovery, Adv. Theory Simul. 5 (2022) 2100565, https://doi.org/10.1002/adts.202100565.
[166] M.M. Cencer, J.S. Moore, R.S. Assary, Machine learning for polymeric materials: an introduction, Polym. Int. 71 (2022) 537–542, https://doi.org/10.1002/pi.6345.
[167] A. Sharma, T. Mukhopadhyay, S.M. Rangappa, S. Siengchin, V. Kushvaha, Advances in computational intelligence of polymer composite materials: machine learning assisted modeling, analysis and design, Arch. Comput. Methods Eng. 29 (2022) 3341–3385, https://doi.org/10.1007/s11831-021-09700-9.
[168] C. Kim, R. Batra, L. Chen, H. Tran, R. Ramprasad, Polymer design using genetic algorithm and machine learning, Comput. Mater. Sci. 186 (2021) 110067, https://doi.org/10.1016/j.commatsci.2020.110067.
[169] H.E. Balcıoğlu, A.Ç. Seçkin, Comparison of machine learning methods and finite element analysis on the fracture behavior of polymer composites, Arch. Appl. Mech. 91 (2021) 223–239, https://doi.org/10.1007/s00419-020-01765-5.
[170] M. Wang, Q. Xu, H. Tang, J. Jiang, Machine learning-enabled prediction and high-throughput screening of polymer membranes for pervaporation separation, ACS Appl. Mater. Interfaces 14 (2022) 8427–8436, https://doi.org/10.1021/acsami.1c22886.
[171] T. Yue, J. He, L. Tao, Y. Li, High-throughput screening and prediction of high modulus of resilience polymers using explainable machine learning, J. Chem. Theory Comput. 19 (2023) 4641–4653, https://doi.org/10.1021/acs.jctc.3c00131.
[172] S. Cassola, M. Duhovic, T. Schmidt, D. May, Machine learning for polymer composites process simulation – a review, Compos. Part B: Eng. 246 (2022) 110208, https://doi.org/10.1016/j.compositesb.2022.110208.
[173] Y. Bai, L. Wilbraham, B.J. Slater, M.A. Zwijnenburg, R.S. Sprick, A.I. Cooper, Accelerated discovery of organic polymer photocatalysts for hydrogen evolution from water through the integration of experiment and theory, J. Am. Chem. Soc. 141 (2019) 9063–9071, https://doi.org/10.1021/jacs.9b03591.
[174] J. Wu, J. Zhang, M. Hu, P. Reiser, L. Torresi, P. Friederich, L. Lahn, O. Kasian, D. M. Guldi, M.E. Pérez-Ojeda, et al., Integrated system built for small-molecule semiconductors via high-throughput approaches, J. Am. Chem. Soc. 145 (2023) 16517–16525, https://doi.org/10.1021/jacs.3c03271.
[175] S. Liu, W. Nie, C. Wang, J. Lu, Z. Qiao, L. Liu, J. Tang, C. Xiao, A. Anandkumar. Multi-modal Molecule Structure-text Model for Text-based Retrieval and Editing. ArXiv abs/2212.10789 (2022).
[176] C.N. Edwards, T. Lai, K. Ros, G. Honke, H. Ji. Translation between Molecules and Natural Language. ArXiv abs/2204.11817 (2022).
[177] G.M. Hocky, A.D. White, Natural language processing models that automate programming will transform chemistry research and teaching, Digit. Discov. 1 (2022) 79–83, https://doi.org/10.1039/d1dd00009h.
[178] A.D. White, G.M. Hocky, H.A. Gandhi, M. Ansari, S. Cox, G.P. Wellawatte, S. Sasmal, Z. Yang, K. Liu, Y. Singh, et al., Assessment of chemistry knowledge in large language models that generate code, Digit. Discov. 2 (2023) 368–376, https://doi.org/10.1039/d2dd00087c.
[179] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. Ponde, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman; et al. Evaluating Large Language Models Trained on Code. ArXiv abs/2107.03374 (2021).
[180] C.M. Castro Nascimento, A.S. Pimentel, Do large language models understand chemistry? A conversation with ChatGPT, J. Chem. Inf. Model. 63 (2023) 1649–1655, https://doi.org/10.1021/acs.jcim.3c00285.
[181] S.H.M. Mehr, M. Craven, A.I. Leonov, G. Keenan, L. Cronin, A universal system for digitization and automatic execution of the chemical synthesis literature, Science 370 (2020) 101–108, https://doi.org/10.1126/science.abc2986.
[182] S. Steiner, J. Wolf, S. Glatzel, A. Andreou, J.M. Granda, G. Keenan, T. Hinkley, G. Aragon-Camarasa, P.J. Kitson, D. Angelone, et al., Organic synthesis in a modular robotic system driven by a chemical programming language, Science 363 (2019) eaav2211, https://doi.org/10.1126/science.aav2211.
[183] N.J. Szymanski, B. Rendy, Y. Fei, R.E. Kumar, T. He, D. Milsted, M.J. McDermott, M. Gallant, E.D. Cubuk, A. Merchant, et al., An autonomous laboratory for the accelerated synthesis of novel materials, Nature (2023), https://doi.org/10.1038/s41586-023-06734-w.
[184] N. Yoshikawa, M. Skreta, K. Darvish, S. Arellano-Rubach, Z. Ji, L. Bjørn Kristensen, A.Z. Li, Y. Zhao, H. Xu, A. Kuramshin, et al., Large language models for chemistry robotics, Auton. Robots 47 (2023) 1057–1086, https://doi.org/10.1007/s10514-023-10136-2.